International Journal of Medicinal Mushrooms, 14(2): 95–134 (2012)

The Role of Culinary-Medicinal Mushrooms on Human Welfare with a Pyramid Model for Human Health

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ABSTRACT: Mushrooms are part of fungal biota characterized by wonder. They rise up from lignocellulosic wastes: yet they become so bountiful and nourishing. Mushrooms are environmentally friendly. They biosynthesize their own food from agricultural crop residues, which would otherwise cause health hazards. The extant records show the continued use of some mushrooms, e.g., *Lentinus edodes*, *Ganoderma lucidum*, and *Cordyceps sinensis* are now centuries old. This review presents a pyramid model for mushroom uses (industries), as food, dietary supplements (tonic), and medicine. A regular intake of mushrooms can make us healthier, fitter, and happier, and help us live longer. The sense of purpose and vision for the mushroom industries is also briefly discussed. A variety of mushrooms have been used traditionally in many different cultures for the maintenance of health and in the prevention and treatment of various diseases. A total of 126 medicinal functions are thought to be produced by medicinal mushrooms (MM) and fungi, including antitumor, immunomodulating, antioxidant, radical scavenging, cardiovascular, anti-hypercholesterolemia, antiviral, antibacterial, anti-parasitic, antifungal, detoxification, hepatoprotective, and anti-diabetic effects. Special attention is paid to mushroom polysaccharides. Many, if not all, higher Basidiomycetes mushrooms contain biologically active polysaccharides in fruit bodies, cultured mycelium, and cultured broth. The data on mushroom polysaccharides are summarized for approximately 700 species of higher Hetero- and Homobasidiomycetes. In particular, the most important for modern medicine are polysaccharides with antitumor and immunostimulating properties. Several of the mushroom polysaccharide compounds have proceeded through phase I, II, and III clinical trials and are used extensively and successfully as drugs in Asia to treat various cancers and other diseases. Mushrooms are superior sources of different types of dietary supplements (DSs) (tonics). The advantages of using mushroom-based DSs as a matter of safety (as opposed to herbal preparations) are: (1) The overwhelming majority of mushrooms used for production of DSs are cultivated commercially (and not gathered in the wild). (2) Mushrooms are easily propagated vegetatively and thus keep to one clone. The mycelium can be stored for a long time, and the genetic and biochemical consistency can be checked after a considerable time. (3) The main advantage, in our opinion, is that many mushrooms are capable of growing in the form of mycelial biomass in submerged cultures. In this review, we discuss legal and regulatory issues introducing and controlling DSs from MMs in different countries, including the United States, the European Community, Australia, New Zealand, Japan, and P.R. China, and guidelines of the World Health Organization. One of the targets of the present review is also to draw attention to many critically important unsolved problems in the future development of medicinal mushroom science in the 21st century.

KEY WORDS: medicinal mushrooms, mushroom industrial uses, dietary supplements (tonics), mushroom medicine, botanical drugs, edible, inedible, hallucinogenic, poisonous, mushroom products, polysaccharides, polysaccharide-protein complexes, beta-glucans, antitumor, immunomodulator activity, macrophages, *Ganoderma lucidum*, *Lentinus edodes*, *Trametes versicolor*, *Cordyceps sinensis*

I. INTRODUCTION

Today the well-being of humankind faces unprecedented challenges involving inadequate regional food supplies, deficiency in new insight into healthy eating, diminishing quality of health, and increasing environmental deterioration. The magnitude of these problems is set to increase as the world's population continues to grow. The 20th century began with a world population of 1.6 billion, and ended with 6.0 billion inhabitants. At the end of October 2011, according to the UN Population Division, there are 7 billion. By 2050 the total world population could reach 10.5 billion, with most of the growth occurring in the less developed countries. With the population still growing by about 80 million each year, it is hard not to be alarmed. Inevitably, the amount

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ABBREVIATIONS: ADoHA: Australian Department of Health and Ageing; **AHCC:** active hexose correlated compound; **AHPA:** American Herbal Products Association; **AIDS:** acquired immune deficiency syndrome; **ANZFA:** Australia New Zealand Food Authority; **BRMs:** biological response modifiers; **CAPE:** caffeic acid phenethyl ester; **COX:** cyclooxigenase; **CRN:** Council for Responsible Nutrition; **Dectin-1:** dendritic-cell-associated C-type lectin-1**; Dectin-2:** DC-associated C-type lectin 2; **DS:** dietary supplement; **DSHEA:** Dietary Supplement Health and Education Act; **EC:** European Commission; **ELISA:** enzyme-linked immunosorbant assay; **EU**: European Union; **FAO:** Food and Agriculture Organization; **FDA:** Food and Drug Administration; **FHC:** Food with Health Claims; **FNFC:** Foods with Nutrient Function Claims; **FOSDU:** Food for Special Dietary Uses; **FSANZ:** Food Standards Australia New Zealand; **FTC:** Federal Trade Commission; **FW:** fresh weight; **GMO:** Genetically Modified Organism; **GMP:** Good Manufacturing Practices; **GRAS:** Generally Recognized as Safe; **HIV:** human immunodeficiency virus; **HPLC:** high-performance liquid chromatography**; IADSA:** International Alliance of Dietary/Food Supplement Associations; **IJMM:** the International Journal of Medicinal Mushrooms; **IMMC:** the International Medicinal Mushroom Conference; **ISMS:** The International Society of Mushroom Science; **IWEMM:** The International Workshops on Edible Mycorrhizal Mushrooms; **IκBa:** inhibitory proteins kappa B; **iNOS:** inducible nitric oxide synthase; **IP:** intellectual properties; **MM:** medicinal mushrooms; **NF-κB:** nuclear factor kappa B**; NK cells:** natural killer cells; **NIH:** National Institutes of Health; **NLEA:** Nutrition Labeling and Education Act; **NNFA:** Nutritional Foods Association; **NZDSR:** New Zealand Dietary Supplements Regulations; **NZFSA:** The New Zealand Food Safety Authority; **ODS:** Office of Dietary Supplements; **OTC:** over the counter; **QHC:** qualified health claims; **PCR:** polymerase chain reaction; **RT-PCR:** reverse transcription PCR; **RDA:** recommended daily allowance; **SF:** "Structure/Function"; **SSA:** significant scientific agreement; **TCM:** traditional Chinese medicine; **TGA:** Therapeutic Goods Administration; **TLC:** thin-layer chromatography; **WHO:** World Health Organization; **WSMBMP:** The World Society for Mushroom Biology and Mushroom Products; **WTO:** World Trade Organization.

of food and the level of medical care available to each individual will decrease, and global ecosystems will be subjected to intensified abuse. It has been reported recently that "global population growth is looming as a bigger threat to the world's food production and water supplies, health quality, than climate change." It has been noted that a nutritious balance of foods and an active lifestyle under a friendly environment can help achieve optimal health throughout life.

We live in an age of human health crises, especially when considering the leading killer diseases of our time. We have the HIV/AIDS pandemic; for an example, Swaziland is today the world's worst HIV/AIDS epidemic: infection among pregnant women is 41%, Swazi life expectancy is down to 43 years, and 31% of Swazi children are orphans.[1] The Black Death (bubonic plague) was a catastrophe of the 14th century (caused by the bacterium *Pasteurella pestis*) that killed 50 million people in Europe and Asia. Today, we have an upsurge of hypertension, diabetes, and cardiovascular disorders worldwide. More than two-thirds of U.S. adults and more than a third of children are overweight or obese.[2] Incidents of cancer (e.g., breast cancer, ovarian cancer, prostate cancer, liver cancer, and various other tumors) are also escalating. A total of 1,596,670 new cancer cases and 571,950 deaths from cancer are projected to occur in the United States in 2011.[3] Mad-cow disease (a viral disorder affecting the nervous system), foot-and-mouth disease (a highly infectious viral problem afflicting cattle, pigs, sheep, and goats, currently causing havoc in Europe), and other disorders (many of which are linked with what we eat) are sending out shock waves that are forcing the world's inhabitants to drastically change their choices of food menus. Human beings are the only species in the world that has figured out how to be in complete control of its food supply. The challenge now is to make sure the food doesn't take control of us.[2]

Medicinal mushrooms (MM) and mushroom products have definitely arrived. They are increasingly becoming a vital component of the human diet throughout the world, as this review will abundantly illustrate. The morphological features and physiological functions of mushrooms may be briefly described by a charming mushroom poem: "Without leaves, without buds, without flowers, yet, they form fruit; as a food, as a tonic, as a medicine, the entire creation is precious."[4] The first clause describes the morphological characteristics of mushrooms, while the second states the physiological functions of mushrooms shown in Fig. 1.

The above pyramid model of mushroom industry conforms fully to an old Chinese saying, "Medicines and food have a common origin." This statement is particularly applicable to mush-

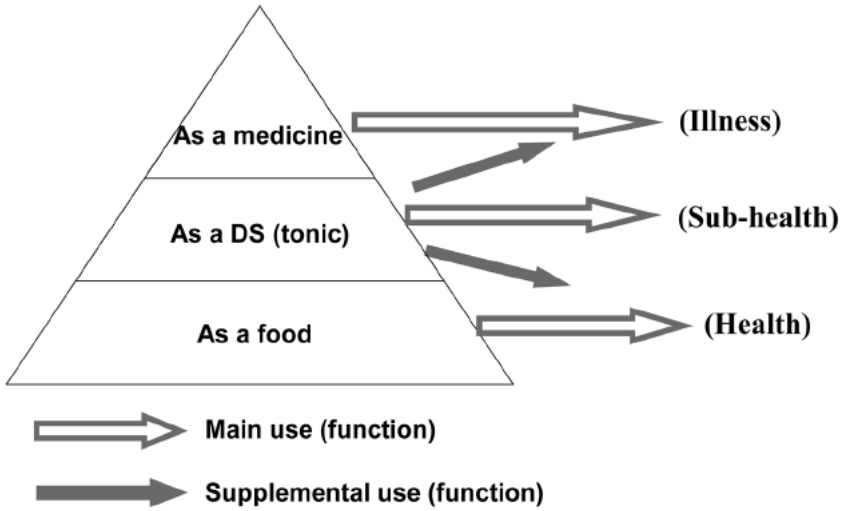

FIGURE 1. A pyramid model of mushroom uses (industry).

rooms, whose nutritional qualities, as well as dietary supplements (DSs) and medicinal attributes, have long been recognized. Diet therapy, the use of foods to sustain or improve health or treat illness, was used by ordinary people and in the imperial court of China as far back as 2000 years ago.[5] Human health may be divided into three states: health, sub-health, and illness. Mushrooms can be mainly used as food for a healthy state, mainly used as a medicine for ill states, and mainly used as DSs for a sub-healthy state as well as for both healthy and ill states (Fig. 1).

It may be useful to reconsider the definition of a mushroom from a global point of view. As we understand, the word *mushroom* may mean different things to different people in various countries. This seems similar to the fable concerning the three blind men and the elephant. Each man developed his own mind picture according to the nature of his particular contact with the elephant. The blind man who touched the sinuous and writhing trunk thought of a snake; he who took hold of the firm tough tail thought of a rope; and he who felt the rough, massive leg pictured the base of a tree.[6] A broad use of the term *mushroom* embraces all large fungi, or all fungi with stalks and caps, or all large fleshy fungi. A more restricted use includes just those larger fungi that are edible and/or medicinal in value. The most extreme use of the term *mushroom* is in reference to just the edible species of genus *Agaricus*. For example, the mushroom industry in the UK and other western countries is nearly 100% dominated by *Agaricus bisporus*. This could lead to the mistaken idea that this is the only species considered as "mushrooms"—some industries even consider brown mushrooms as exotic![7] Actually, there are thousands of different speciess of mushrooms. It is understood that mushrooms are generally considered as a group of macrofungi, which have larger fruit bodies. According to the definition given by Chang and Miles,[8] a mushroom is a "macrofungus with a distinctive fruiting body which can be either epigenous or hypogeous and large enough to be seen with the naked eye and to be picked by hand." Therefore, mushrooms need not be Basidiomycetes, nor aerial, nor fleshy, nor edible. Mushrooms can be Ascomycetes as well. In other words, mushrooms can be roughly divided into six categories: edible, inedible, hallucinogenic, medicinal, poisonous, and a miscellaneous category whose properties remain less well defined. This definition is imperfect, because it is inevitably loose and subjective; however, so far, no alternative definitions are proposed.[9] Therefore, it can be used as a working term for global mushrooms in both mushroom biology and the mushroom industry. It is believed that this simple definition is better than the terms *edible fungi* and *toadstool* used in some countries to mean edible and poisonous mushrooms, respectively. This is because mushrooms are macrofungi (thereby excluding microfungi); however, the term *edible fungi* can include both macro-and microfungi.

The term *toadstool* is meaningless and should be avoided to prevent confusion.

Mushrooms have been found in fossilized wood 300 million years old.[10] Almost certainly, prehistoric man used mushrooms collected in the wild as food. There is ample evidence that the great early civilizations of the Greeks, Egyptians, Romans, Chinese, and Mexicans prized mushrooms as a delicacy, for their purported therapeutic values, and, in some cases, as treasures in religious rites. Throughout recorded history there are repeated references to the use of mushrooms as food and for medicinal purposes, and it is not surprising that the intentional cultivation of mushrooms had a very early beginning. China can boast that it was the first to successfully cultivate many popular mushrooms species—for example, *Auricularia auricula-judae* (estimated date, 600 AD), *Flammulina velutipes* (800–900 AD), *Lentinus edodes* (1000–1100 AD), *Volvariella volvacea* (1700 AD), and *Tremella fuciformis* (1800 AD). Prior to the 1900s, *Agaricus bisporus* (1650 AD in France) was the only major commercially cultivated mushroom species that was not first cultivated in China.[4]

Knowledge of numerous new mushroom species has accumulated through time. Out of the 1.5 million estimated fungi species, Hawksworth[9,11–13] estimated that 140,000 species produce fruiting bodies of sufficient size and suitable structure to be considered as macrofungi. These can be called mushrooms according to the above definition. It should be noted that without a globally recognized term for mushrooms, it is impossible to make an estimation of the number of so-called mushrooms on the earth. The number of recognized mushroom species has been reported to be 14,000,[9,11–13] which account for 10% of the estimated 140,000 mushroom species. Therefore, there are currently still many unknown species to be discovered (about 126,000). Of the recognized mushroom species, about 7,000 of the species (50%) are considered to possess varying degrees of edibility, and more than 3,000 species from 231 genera are regarded as prime edible mushrooms. But only about 200 of the prime edible mushrooms are experimentally grown, 100 economically cultivated, around 60 commercially cultivated, and more than 10 produced on an industrial scale in many countries. Furthermore, of the 14,000 known mushroom species, approx 700 are considered to be safe species with medicinal properties.[14] The number of poisonous mushrooms approximates 500 species. China is estimated to have about 2,000 mushroom species, with 981 known to be edible species. By 2002, 92 species had been domesticated, while 50 of these had been commercially cultivated.[15]

The outlook for many of these known mushroom species is bright. The current production level stands at 24 million tonnes of edible mushrooms annually. The world market for the three main categories in the mushroom industry (edible mushrooms, mushroom products, and wild mushrooms) is valued at over US$60 billion. Furthermore, the production of mushrooms worldwide has been steadily increasing, mainly due to contributions from developing countries, such as China, India, Poland, Hungary, and Vietnam. In addition, technological developments in the mushroom industry have seen increasing production capacities, innovations in cultivation technologies, improvements to final mushroom goods, and utilizing of the mushroom's natural qualities for environmental benefits. However, there is always the need to maintain current trends and to continue to seek out new opportunities. Increasing consumption capabilities arising from an increasing world population should be recognized. This opportunity for the mushroom industry should be grasped by encouraging the increase of mushroom consumption, through active promotion of mushroom properties and new advantages of modern marketing techniques. Certainly, quality control is of paramount importance in securing and attracting further consumer confidence in the mushroom industry. We hope and believe that this review will serve as a catalyst for promoting the increased use of mushrooms in human diets; for promoting research and development in world mushroom biota; and for looking at mushrooms as one of the possible sources of solutions for some of the health problems of our time.

II. WORLD MUSHROM PRODUCTION

Since the end of World War II, mushroom production has steadily increased in agricultural-based industries. World production of cultivated edible mushrooms over a number of years is shown in Table 1. In 1981, the production totalled 1,257.2 thousand tonnes; in 1986, 2,182.0 thousand tonnes; in those five years, mushroom produc-

TABLE 1. World Production of Cultivated Edible Mushrooms (Fresh Equivalent, from 1960 to 2009)

Year	Production (´1,000 MT)	Increase %	Annual increase %
1960	170.0		
1965	341.0	100.6	20.1
1970	546.0	60.0	12.0
1975	916.0	67.8	13.6
1981	1,257.2	37.3	6.2
1986	2,176.0	73.1	14.6
1990	3,794.0	74.4	18.6
1991	4,273.0	12.6	12.6
1994	4,909.3	14.9	5.0
1997	6,158.4	25.4	8.5
2002	12,250.0	98.9	19.8
2009	24,000.0*	95.9	13.7

Annual average increase 13.2%

Sources: Delcaire,[16] Sharma,[17] Chang,[18] [* this figure assumes that Chinese contribution was 20.20 million tons (Li[19]) and the rest was made up by all other countries].

tion increased by 73.6% and there was an annual increase of 14.7%; in 1990, 3,763.0 thousand tonnes; in those four years, production increased 72.5%, with annual increases of 18.1%; in 1994, 4,909.3 thousand tonnes was produced; in those four years, production increased by 30.5%, with an annual increase of 7.6%; and in 1997, production totalled 6,158.4 thousand tonnes; in those three years, production increased 25.4%, with an annual increase of 8.5%. Overall, the world mushroom production increased over 12% annually during the period from 1981 to 1997. However, the *Agaricus* mushrooms decreased in percentage of world total production. This is mainly due to other alternative edible mushrooms becoming more in demand, e.g, *Lentinus edodes* mushrooms increased both in percentage, from 14.3% in 1981 to 25.2% in 1997, and in production, from 180.0 thousand tonnes to 1,564.4 thousand tonnes. *Pleurotus* mushrooms increased from 2.8% to 14.2%, and their production from 35.0 thousand tonnes to 875.6 thousand tonnes, with a 25-fold increase. And *Auricularia* mushrooms increased from 0.8% in 1981 to 7.9 % in 1997, and their production increased from 10.0 thousand tonnes in 1981 to 485.6 thousand tonnes in 1997, with a 48.5-fold increase. Overall, world mushroom production has steadily increased (Table 1), due mainly to contributions from countries with developing economies, including China, India, Poland, and Hungary. In contrast, mushroom production in Western European countries, the United States, and Japan, has remained unchanged or even fallen. China especially has witnessed a huge increase in edible mushroom cultivation, and now makes the largest contribution, by over 80%, to the total worldwide output (Table 2). Furthermore, several new species of mushrooms have been recently cultivated and marketed in China (Table 3). India's annual production of mushrooms doubled from 5,000 tonnes in 2001 to 10,000 tonnes in 2004, and is expected to continue rising, at about 25% per annum, for the foreseeable future.

In Latin America, the annual mushroom production has also increased steadily since 1995. During the period 1995–2001, the estimated commercial mushroom production level in this region rose by 32% (49,975 to 65,951 tonnes), equivalent to an annual increase of 5%.

Since mushroom cultivation can be a labor-intensive agro-industrial activity, it could have great economic and social impact by generating income and employment for both women and youth, particularly in rural areas in developing countries. Using China as an example, in 1978 the total production of mushrooms in China was only 60,000 tonnes, which accounted for less than 6% of total world mushroom production. In 2009, however, total production of mushrooms in China reached 20.2 million

TABLE 2. China's Contribution to Worldwide Mushroom Production Since 1978

Year	Total production (˜1,000 MT)	China's production (˜1,000 MT)	China's contribution (%)
1978	1,060.0	60.0	5.7
1983	1,453.0	174.5	12.0
1990	3,763.0	1,083.0	28.8
1994	4,909.3	2,640.0	53.8
1997	6,158.4	3,918.0	63.6
2002	12,250.0	8,650.0	70.6
2006	NA	14,000.0	NA
2009	24,000	20,200.0	More than 80

Sources: Chang,[20,21] Huang,[22] and courtesy communications with the China Edible Fungi Association (2011).[23.] NA: not available.

tonnes and accounted for over 80% of total world mushroom production. According to recent statistics, in 2009 the value of total mushroom production in China was US$20 billion and the export value of mushrooms was US$1.3 billion. It is estimated that in 2010, the total production will be 22 million tonnes and the total value will be about US$23 billion; the export value of mushrooms could be US$2 billion. The export value accounts for only about 10% of the total production value. Total employment in the mushroom industry in China was over 30 million people in 2006, with only 10% of them employed as actual mushroom farmers; other employment categories were in sectors such as food,

TABLE 3. New Species of Culinary-Medicinal Mushrooms Cultivated Recently in China, MT

Year	*Flammulina velutipes*	*Tremella* spp.	*Pholiota nameko*	*Coprinus comatus*	*Dictyophora* spp.
1986	10	50	—	—	—
1998	189	100	31	—	—
2001	389	114	51	39	10
2003	557	183	172	178	32
2005	838,517	184,130	203,746	285,100	37,464
2007	1,177,962	251,414	259,175	441,869	77,710
Year	***Hypsizygus* spp.**	***Pleurotus nebrodensis***	***Pleurotus eryngii***	***Agrocybe chaxingu***	***Agaricus brasiliensis***
1986	—	—	—	—	—
1998	21	—	—	—	—
2001	120	7	21	—	—
2003	243	52	114	93	42
2005	346,426	172,727	135,943	191,346	43,607
2007	421,406	181,671	202,302	232,868	47,030
Year	***Volvariella volvacea***	***Hericium erinaceus***	***Pleurotus cornucopiae***	***Pleurotus geesteranus***	***Grifola frondosa***
1986	38	—	—	—	—
1998	32	28	—	—	10
2001	116	10	—	—	15
2003	174	31	—	—	25
2005	274,338	45,348	68,038	95,762	4,875
2007	437,256	57,231	102,726	133,596	7,115

Source: Courtesy communications with the China Edible Fungi Association.[24]

beverage manufacturing, trading and management, transport, marketing, wholesaling, retailing, export, etc. The mushroom industry can also have even broader positive spillover, generating complementary employment in areas such as accommodation, restaurant services, etc. Furthermore, it is interesting to note that in some counties in China, with a population of just under 200,000 people, 60% of the population were engaged in mushroom production and management. The local mushroom industry can also be the main source of revenue for local government. The development of the mushroom industry in China can be used as a model for other less-developed countries.

Due to the uneven development of regional economies, Chinese mushroom development models/methods and market development have become diversified. Around the metropolises like Beijing, Shanghai, and Shenzhen, rising numbers of big and modern exotic mushroom farms/factories emerged, while the number of traditional family-run farms began to decline. City dwellers preferred fresh exotic mushrooms, e.g., fresh enokitake (*Flammulina velutipes*), buna shemeji (*Hypsizygus marmoreus)*, and king oyster (*Pleurotus eryngii*). In general, a modern farm with a bottle system could produce 5–10 tonnes of fresh mushrooms per day. Several of the modern farms in China produce more than 40 tonnes of fresh mushrooms daily. More than 90% of the production is sold in domestic markets. Actually, about ten years ago, China started to import mushrooms. Mushroom imports in China hit a new record in 2009. The turnover was more than US$6 million, which was three times that of 2008. Korean fresh *Flammulina velutipes* accounted for 70% of total imports. Both fresh and canned *Volvariella volvacea* are also imported in considerable quantity from Vietnam.

This dramatic expansion in mushroom production in China in the last 12 years (1990–2009) from 1 million tonnes to 20.2 million tonnes is mainly due to:

(1) The strong leadership and initiative of provincial and county governments in grasping cultivation techniques and promoting production. Under these governments, there is usually a special section called "Office of Mushroom Production," which bears responsibility for mushroom research and production and training. The office frequently organizes mushroom farming training courses for farmers.

(2) The strong scientific support with the proliferation of mushroom scientists and scholars from various academic institutions in China. It has been estimated that there are more than 500 postgraduate students working for their MSc and PhD degrees in different fields of mushroom biology and applied mushroom biology.[25]

(3) The great number of new mushroom species (varieties), both edible and medicinal, that have been domesticated and commercialized, e.g., *Pleurotus eryngii*, *P. ferulae*, *P. nebrodensis*, *P. abalonus*, *P. pulmonarius*, *P. corucopiae*, *P. tuber-regium*, *Flammulina velutipes*, *Coprinus comatus*, *Agrocybe chaxingu*, *Dictophora duplicata*, *Hypsizygus marmoreus*, *Macrocybe lobayensis*, *Hericium erinaceus* mainly for edible, and *Phellinus linteus*, *Antrodia cinnaomea* (=*Taiwanofungus camphorates* (niuzhangzhih in Chinese), *Cordyceps militaris*, *Xylaria nigripes* (wulingshen in Chinese), *Phellinus baumii* (sanghuang in Chinese), mainly for medicinal use.

(4) The many innovations of mushroom cultivation technology by talented mushroom farmers. The synthetic log method for cultivation of *Lentinus edodes* was invented by farmer Pan Zhaowan in Gutian County in the Province of Fujian. This has raised the living standards of the farmers.

(5) The increase of the domestic market due to the economic boom for the last three decades, which is also a key factor in the expanding cultivation of edible and medicinal mushrooms in China. A survey conducted recently found that mushroom production and product industries are turning their attention toward the rising economies of Asia, and particularly of China.

Production patterns vary according to the geographic region. Although *Agaricus bisporus* is only one of the many edible mushrooms cultivated globally, it has remained the dominant cultivated mushroom in countries of the Western world. In the United Kingdom and most other Western European states, the mushroom industry is overwhelmingly focused on this species. This production pattern towards *A. bisporus* is slightly less biased in Spain, the third largest mushroom producer in the European Union countries. *A. bisporus* accounts for 80% of mushroom production in Spain, where total production rose to 110,000 tonnes in 2004, compared with 26,512 tonnes in 1992 (a 315% increase). In the United

TABLE 4. Relative Contributions of *Agaricus bisporus*, *Lentinus edodes*, and *Pleurotus* spp. to the Mushroom Industry of the United States

Species	Production (MT)		
	2002–03	2003–04	2004–05
Agaricus bisporus	379318.8	381479.4	380083.0
	(98.6%)	(98.4%)	(98.2%)
Lentinus edodes	3390.5	3506.1	4118.4
	(0.9%)	(0.9%)	(1.1%)
Pleurotus spp.	1812.6	2008.2	2453.0
	(0.5%)	(0.5%)	(0.6%)
Sub-total	384521.9	386993.7	386654.4
	(99.9%)	(99.9%)	(99.9%)
Others	197.3	541.0	253.5
	(0.01%)	(0.01%)	(0.01%)
Grand total	384719.2	387534.7	386907.9
	(100%)	(100%)	(100%)

Source: National Agricultural Statistics Service (NASS),[26] USA.
Note: Share of total US mushroom production is shown in parentheses.

States, about 98% of mushroom production is also *A. bisporus*. However, in 2003, *A. bisporus* accounted for only 12.8% and 11.6% of the total mushroom production in China and South Korea, respectively, and 0% in Japan.

In contrast, mushroom species other than *A. bisporus* are far more popular in East Asian countries. This applies especially to *Lentinus edodes* and species of genus *Pleurotus*, which, based on recent production figures, have already become the two most popular mushrooms grown worldwide. It should be noted that mushrooms long held popular in Asia (e.g., *Lentinus edodes*, *Flammulina velutipes*, and *Pleurotus* spp.) are now making inroads into Western markets. In Spain, *Pleurotus* spp. and *L. edodes* make up 15% and 5%, respectively, of total mushroom production, although this compares with corresponding figures of only 0.5% and 1% in the United States (Table 4). However, whereas *Agaricus*, *Lentinus*, and *Pleurotus* mushrooms together make up nearly 100% of the mushroom production in the U.S and Spain, these three mushrooms account for 72.7%, 58.2%, and 12.3% of the total mushroom output of South Korea, China, and Japan, respectively (Table 5). Clearly, a much wider range of other culinary and medicinal mushrooms are cultivated and marketed in these three Asian countries, particularly in Japan. Although very few of these so-called specialty mushrooms have so far been introduced into Western markets in recent years, they will undoubtedly become increasingly more important in mushroom markets worldwide.

III. TOTAL UTILIZATION OF LIGNOCELLULOSIC BY-PRODUCTS (WASTE MATERIALS) BASED ON MUSHROOMS

As mentioned, the population of the world is expected to continue increasing in the 21st century, so the amount of food and the level of medical care available to each individual, especially those living in less developed countries, will decrease. Environmental pollution will also become a more serious problem. However, the world has an immense amount of lignocellulosic by-product resources, which, like solar energy, is sustainable. This resource has been considered insignificant or of no commercial and certainly no food value, at least in their original forms. It should be noted that significant research funds have been set aside to search for increased productivity in core products, such as the kernel of maize, the grain of the cereal crop, the oil in coconut, the cellulose in trees, the fiber in sisal, and coffee in the coffee cherry. However, few research funds have been reserved to search for the possible reuse of many by-products, e.g., cereal straw, coconut coil, spent coffee grounds, etc., which are usually considered waste materials. This concept and practice have been called[18] linear thinking on core products of agricultural industry in the 20th century. This means that the by-products from processing

TABLE 5. Relative Contributions of *Agaricus bisporus*, *Lentinus edodes*, and *Pleurotus* spp. to the Mushroom Industry in Asia

Species	Production (1000 MT)			
	China	Japan	S. Korea	Taiwan
Agaricus bisporus	1330.4	—	19.790	4.276
	(12.8%)		(11.6%)	(4.0%)
Lentinus edodes	2228	35.294	41.876	36.000
	(21.5%)	(10.7%)	(24.6%)	(33.4%)
Pleurotus spp.	2488	5.219	61.965	4.540
	(24.0%)	(1.6%)	(36.5%)	(4.2%)
Sub-total	6046.4	40.513	12.3631	44.816
	(58.2%)	(12.3%)	(72.7%)	(41.6%)
Others	4340.5	290.333	46.369	62.984
	(41.8%)	(87.8%)	(27.3%)	(58.4%)
Grand total	10386.9	330.846	170.000	107.800

Source: Cui,[27] Ho, Peng,[28] courtesy communications with K. Yamanaka. Note: Shares of total Asia mushroom production shown in parentheses.

of core products are usually considered as wastes and are not properly reused (Fig. 2). When they are carelessly dispensed to the surrounding environment by dumping or burning, these so-called wastes are bound to lead to environmental pollution and consequent health hazards. It should be emphasized that these lignocellulosic wastes are actually a kind of natural resource or raw material; if they could be properly managed and utilized, then eventually greater economic growth would be possible. The second approach is cluster thinking, which should be applied to agriculture-based industry in the 21st century (Fig. 3). In other words, the by-products in processing core products should be reused/treated as raw materials for the production of secondary or tertiary core products; such materials as cereal straw, coffee pulp, spent coffee grounds, sisal waste, all can be used to grow mushrooms. After harvesting mushrooms, the spent compost can be used as feed materials for animals or for growing earthworms. Afterward, the residues can be used as soil conditioners or crop fertilizers. In the whole exercise, there is no waste produced. This is the concept of zero emissions or total productivity.[29] This approach conceptualizes the conversion of biota by-products, which are generally viewed as wastes, into resources, thereby creating employment, reducing poverty, and generating opportunities in affected areas.

Mushrooms, like all other fungi, lack chlorophyll and are non-green organisms. They cannot convert solar energy through the process of photosynthesis to organic matter, as green plants do, but they can produce extensive enzymes, which can degrade lignocellulosic materials as nutrition for

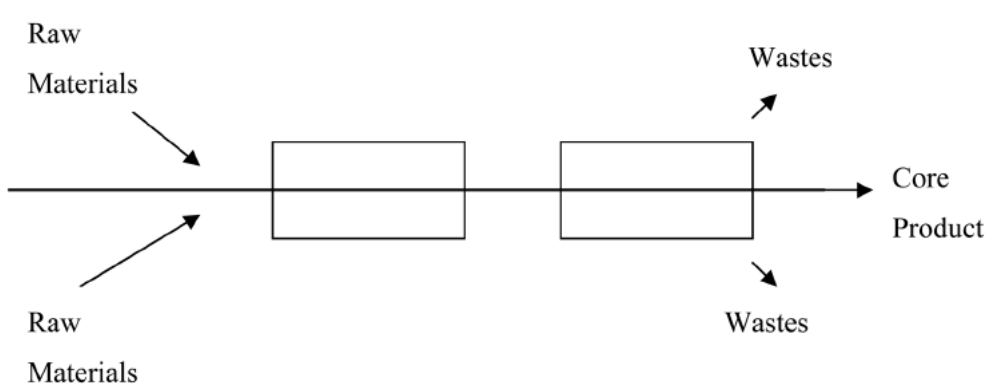

FIGURE 2. 20th century—Linear thought on agricultural/mushroom-based industry.

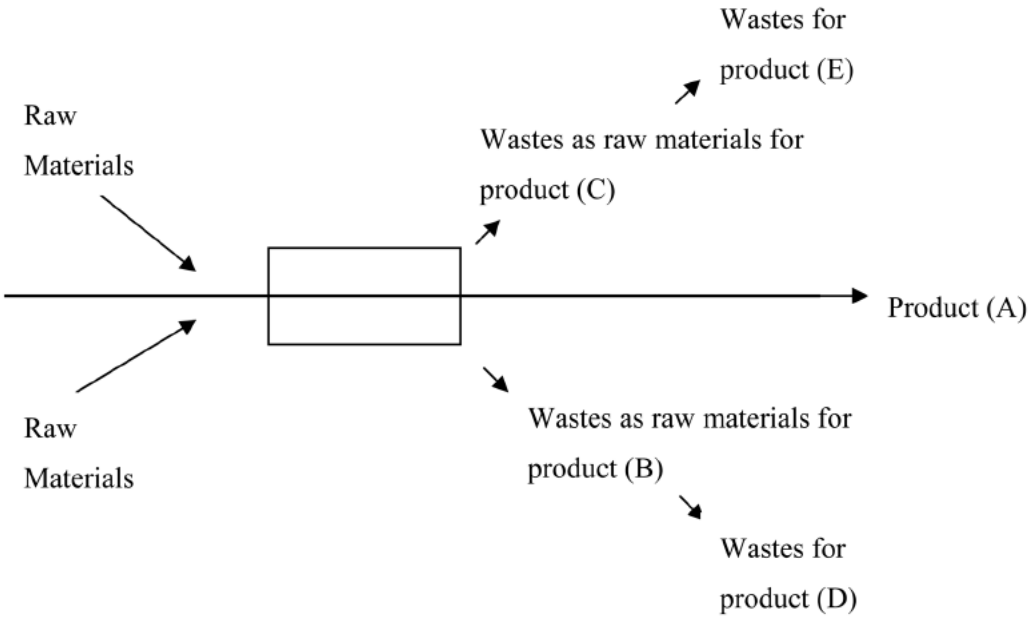

FIGURE 3. 21st century—Cluster thought on agricultural/mushroom-based industry.

their growth and fruiting. This is to demonstrate the impressive capacities of mushrooms for biosynthesis, which is different from photosynthesis by green plants. Mushrooms not only can become nutritious protein-rich food, but also can be served as nutriceuticals or even as pharmaceutical products. Mushrooms have great potential for domestic consumption, as well as for export, providing international quality standards and timely supply schedules can be maintained. The most significant aspect, according to the cluster approach to mushroom-based industry, is to recycle the by-products (wastes) in the course of each stage of mushroom production and to create a pollution-free environment. By blending the advance in basic biological knowledge with that of practical technology, a mushroom-related industry based on utilization of the lignocellulosic waste materials that are abundantly available in rural and urban areas can have a positive global impact on long-term food nutrition, health enhancement, environmental conservation and regeneration, and economic and social benefits. Therefore, the significant impact of mushroom cultivation and mushroom derivatives/products on human welfare in the 21st century could be considered globally as a non-green revolution,[18] but it must be implemented according to locally available substrates, labor, and climatic conditions.

IV. DEVELOPMENT OF WORLD MUSHROOM INDUSTRY

Before the 1st International Conference on Mushroom Biology and Mushroom Products in 1993, the international mushroom movement/industry concentrated mainly on the production of fresh, canned, and dried mushrooms consumed for food, valued at US$8.5 billion in 1991. This was considered to be only one leg of the mushroom industry. However, US$1.2 billion was estimated to have been generated from MMs (mushroom products) in the same year, which did not receive prominent attention. During that conference, the MMs and their products used as tonic and dietary supplements/herbal medicines had been promoted and were considered to be the second leg of the mushroom industry. Since then, the MM movement has rapidly expanded worldwide. This means the second leg of the mushroom industry has been growing stronger and stronger. The world market for edible mushrooms was estimated in 2006 at around US$36 billion, and MM products were valued at US$16 billion.

There are four big international bodies organizations/forums which target each of these segments of the mushroom industry, and they have subsequently evolved and have served to bring their respective divisions to the forefront of international attention.[25] The International Society

of Mushroom Science (ISMS), focused on edible mushrooms, was established in 1950 in England. Since then, ISMS congresses have been held every 3–4 years, and the 18th ISMS Congress will be held in 2012 in Beijing, China. The World Society for Mushroom Biology and Mushroom Products (WSMBMP), focusing on mushroom biology and MM products, was formed in 1993 in Hong Kong. The WSMBMP hosts an international conference every three years; the 7th International Conference on Mushroom Biology and Mushroom Products was held in 2011 in France. The International Workshops on Edible Mycorrhizal Mushrooms (IWEMM), which relate primarily to wild mushrooms, were inaugurated in 1999 in Sweden. These workshops are held every two years. The IWEMM-6 was held in Morocco in 2011. The international movement for the medicinal segment of the mushroom industry received a further boost following the launch in 1999 of the International Journal of Medicinal Mushrooms (IJMM). Two years later, in 2001, the inaugural International Medicinal Mushroom Conference (IMMC) was held in Kiev, Ukraine. These conferences are held at two-year intervals, with IMMC-7 scheduled for 2013 in China. Compared to a four-legged horse, these four international mushroom organizations have carried the world mushroom industry forward. They have each done much to promote their respective areas of primary focus, not least by bringing together scientists and growers for mutually informative interactions at a personal level, by encouraging research, and by providing platforms for the dissemination of valuable research and commercial information. The rider represents the scene of mushroom consumers, who are the driving force for expanding mushroom markets and for enhancing the mushroom research and mushroom industry development.

V. MUSHROOMS AS A FOOD

Although mushrooms have been used as a staple food in the diet of some cultures and were also known to provide nutrition during times of famine, edible mushrooms are usually considered only for their flavor and condiment value. Varying opinions have been expressed regarding the true nutritive value of edible mushrooms.[30] Since the early 1970s, a considerable number of definitive studies conducted by competent nutritionists and mushroom scientists have become available. Most of the data on the nutritional contribution as well as medicinal effects of mushrooms to the human diet and health have been published. The true nutritive value of mushrooms has rapidly become known and recognized not only by mushroom researchers and farmers but also by the general public. Experimental data have been presented at international conferences, particular in the four big mushroom bodies/organizations as mentioned above, and scientific papers published in various international scientific journals, which all demonstrated that edible mushrooms contain not only high nutritional values but also show medicinal effects. Several comprehensive reviews of the nutritional value of mushrooms have been presented.[30–32]

The moisture content of fresh edible mushrooms varies from 70% to 95%, depending upon the harvest time and environmental conditions, whereas it is only about 10% to 13% in dried mushrooms. The protein content of the cultivated edible species ranges from 1.75% to 5.9% of their fresh weight. It has been estimated that an average value of 3.5%–4% would be more representative. This means that the protein content of edible mushrooms, in general, is about twice that of the onion (1.4%) and cabbage (1.4%), and four times and twelve times those of oranges (1.0%) and apples (0.3%), respectively. In comparison, the protein content of common meats is as follows: pork, 9%–16%; beef, 12%–20%; chicken, 18%–20%; fish, 18%–20%; and milk, 25.2%. On a dry-weight basis, mushrooms normally contain 19%–35% protein, as compared to 7.3% in rice, 12.7% in wheat, 38.1% in soybean, and 9.4% in corn. Therefore, in terms of the amount of crude protein, mushrooms rank below animal meats, but well above most other foods, including milk, which is an animal product. Furthermore, mushroom protein contains all nine essential amino acids required by humans. In addition to their good protein, mushrooms are also a relatively good source of the following individual nutrients: fat, phosphorus, iron, and vitamins including thiamine, riboflavin, ascorbic acid, ergosterine, and niacin. They are low in calories, carbohydrates, and calcium.[4] It has been reported[33] that a total lipid content varying between 0.6% and 3.1% of dry weight was found in the commonly cultivated mushrooms. At least 72% of the total fatty acids were found to be

unsaturated in all these mushrooms. Unsaturated fatty acids are essential and significant in our diet and to our health. Mushrooms have a nucleic acid content low enough to allow regular consumption as a form of vegetable.[34] Generally, edible mushrooms possess all three properties of food—nutrition, taste, and physiological functions.

For the first property (i.e., nutritional value), mushrooms are: rich in proteins, chitin (dietary fibers), vitamins, and minerals; low in total fat but with a high proportion of unsaturated fatty acids, and have no cholesterols. Regarding the beneficial nutritional effects of mushrooms, the following facts should be noted:

Mushrooms have a low energy level, which is beneficial for weight reduction.

Mushrooms have significant levels of purine, which is beneficial for the diet of persons suffering from metabolic diseases (e.g., gout, rheumatism).

Mushrooms have a low glucose level, and more mannitol, which is especially beneficial for diabetics.

Mushrooms have a very low sodium concentration, which is beneficial for the diet of persons suffering from high blood pressure.

Mushrooms have a high content of several key vitamins, which is an important orthomolecular aspect. This means that a significant part of the daily requirement of different essential vitamins can be covered by consuming mushrooms.

Mushrooms have a high content of potassium and phosphorus, which is an important orthomolecular aspect as well.

Finally, mushrooms have a high content of selenium, which is regarded as an excellent antioxidant.[35]

For the second property, taste, mushrooms serve as a delicious foodstuff, and also as a source of food flavoring substances (because of their unique flavors). In addition to the volatile eight-carbon compounds, the typical mushroom flavor consists of water-soluble taste components such as soluble sugars, polyols, organic acids, free amino acids, and free 5′-nucleotides.

For the third property (i.e., of physiological functions), mushrooms contain several physiologically active substances, including high-molecular-weight polysaccharides (mainly β-D-glucans), heteroglucans, chitinous substances, peptidoglucans, proteoglucans, lectins, RNA components, dietary fiber; and low-molecular-weight organic substances, such as terpenoids, steroids, and novel phenols. Therefore, overall, mushrooms are a valuable health food in modern society.

As for tonic and herbal medicinal values, there is a history of empirical observations relating to the beneficial effects of mushrooms on a wide range of human ailments. In recent years, there has been a huge upsurge of interest in mushrooms as a source of new compounds capable of improving biological functions, and in activities related to the use of mushroom products as dietary supplements and for other medicinal purposes.[36] Moreover, the application of modern analytical techniques provides scientific data in support of earlier empirical observations. There is growing experimentally based evidence to suggest that the consumption of mushrooms as a functional food, or the use of extracted bioactive compounds as dietary supplements, increases resistance to disease and, in some cases, causes regression of a diseased state. In many cases, these practices appear to enhance the immune response of the human body, which is thought to be weakened as a result of exposure to increasing levels of stress caused by present-day high-pressure work demands. The term *mushroom nutriceuticals* has been coined to embody both the nutritional and medicinal features of these biological response modifiers extractable from either the mushroom mycelium and/or mycelium culture fluids, or mushroom fruiting bodies.[37] Currently, mushrooms represent one of the most rapidly growing sources of extractables both for inclusion in food supplements to enhance health and fitness, and in the prevention and treatment of human disease conditions. Since mushrooms have a long tradition as a food source, mushroom feedstocks are categorized as "generally recognized as safe." Furthermore, mushroom nutriceuticals are reported to have extraordinary low toxicity, even at high doses, and are apparently lacking in various side effects that frequently accompany the use of synthetic drugs.

There is increasing experimentally based evidence to support centuries of observations regarding the nutritional (and medicinal) benefits of mushrooms. Mushrooms are of high quality and high economic value. The three most popularly cultivated mushrooms, *Pleurotus* spp., *Lentinus edodes*, and *Agaricus bisporus*, are used in

TABLE 6. Proximate Composition of the Three Mushroom Species[36]

Proximate composition	*Agaricus bisporus*	*Lentinus edodes*	*Pleurotus* spp.
Moisture	78.3–90.5	90.0–91.8	73.7–90.8
Crude protein	23.9–34.8	13.4–17.5	10.5–30.4
Crude fat	1.7–8.0	4.9–8.0	1.6–2.2
Total carbohydrate	51.3–62.5	67.5–78.0	57.6–81.8
N-free carbohydrate	44.0–53.5	59.5–70.7	48.9–74.3
Total dietary fiber	8.0–10.4	7.3–8.0	7.5–8.7
Ash	7.7–12.0	3.7–7.0	6.1–9.8
Energy value	328–368	387–392	345–367

All data are presented as percentage of dry weight, except moisture (percentage of fresh weight) and energy value (Kcal per 100 g dry weight).

this section to illustrate the nature of mushrooms as a food. Generally, edible mushrooms possess all three properties of food—nutrition, taste, and physiological functions.

Mushrooms are rich in proteins (Table 6), with all nine essential amino acids (Table 7), chitin, and vitamins, particularly B and D; low in fat with high proportions of unsaturated fatty acids (Table 8) and also low in calories and cholesterols; they have high levels of special minerals such as potassium, phosphorus, and selenium (an excellent antioxidant), but they are low in sodium. A comparison of chemical composition of fruiting bodies and submerged cultivated mycelia of the same species of culinary mushrooms is very important. For example, the study of fruit bodies and mycelia of *Coprinus comatus* was done by Tsai et al.[38] They found that differences in the profile of proximate composition in studied material. Contents of soluble sugars and polyols in fruit bodies and mycelia were 250.82 and 34.45 mg/g, respectively. Trehalose content was the highest in fruit bodies (169.1 mg/g) whereas sucrose content was the highest in mycelia (24.48 mg/g). Contents of total free amino acids in fruit bodies and mycelia were 9.94 and 6.05 mg/g, respectively. The content of monosodium glutamate–like components in fruit bodies (3.23 mg/g) of *C. comatus* was significantly higher than that in mycelia (0.50 mg/g)[38] (Tables 9–11).

The study of Ulziijard and Mau[39] summarizes the results of proximate composition of 38 fruiting bodies and 19 mycelia of 32 species of culinary-medicinal mushrooms from genera *Agaricus*, *Agrocybe*, *Antrodia*, *Auricularia*, *Boletus*, *Clitocybe*, *Coprinus*, *Cordyceps*, *Trametes*, *Dictyophora*, *Flammulina*, *Ganoderma*, *Grifola*, *Hericium*, *Hypsizygus*, *Inonotus*, *Lentinus*, *Morchella*, *Pleurotus*, *Sparassis*, *Termitomyces*, *Tremella*, and *Tricholoma*. Based on the proximate composition, most fruiting bodies and mycelia are low in fat and rich in protein and dietary fiber (DF); however, some are rich in soluble polysaccharides and others are rich in crude fiber (Tables

TABLE 7. Essential Amino acid Contents of the Mushrooms*[35]

Essential amino acids	*Agarius bisporus*	*Lentinus edodes*	*Pleurotus* spp.
Cystine	23	24	28
Methionine	33	29	35
Threonine	111	98	106
Valine	121	124	112
Isoleucine	91	79	82
Leucine	153	133	139
Lysine	143	122	126
Tyrosine	283	265	219
Phenylalanine	107	91	111

*mg/100 g FW

TABLE 8. Fatty Acid Content of the Mushrooms[33]

Species	% of saturated fatty acids	% of unsaturated fatty acids
Agaricus bisporus	19.5	80.5
Pleurotus spp.	20.7	79.3
Lentinus edodes	19.9	80.1

12 and 13). Due to the high amount of DF present, the energy provided by 100 g of dry fruiting bodies and mycelia is 46.96 to 292.37 kcal and 195.84 to 373.22 kcal, respectively. The energy (100 g) is classified into four levels: first level of >300 kcal, second level of 200–300 kcal, third level of 100–200 kcal, and fourth level of <100 kcal. Most fruiting bodies are listed in the third level, nine mycelia are listed in the first level, and ten in the second level. Overall, the information about the proximate composition and energy are of great interest for fruiting bodies and mycelia to be used as foods or food-flavoring materials or in the formulation of health foods.

Mushrooms also serve as delicious foodstuffs and as food flavoring substances because of their unique flavors. In addition to the volatile eight-carbon compounds, the typical mushroom flavor consists of water-soluble taste components such as soluble sugars, polyols, organic acids, free amino acids, and 5′-nucleotides.

Mushrooms contain several physiologically active substances, including high-molecular-weight polysaccharides (mainly β-D-glucans), heteroglucans, chitinous substances, peptidoglucans, proteoglucans, lectins, RNA components, and dietary fiber, as well as low-molecular-weight organic substances, such as terpenoids, steroids, and novel phenols. Therefore, overall, mushrooms are a valuable health food in modern society.

VI. MUSHROOMS AS DIETARY SUPPLEMENTS (TONICS)

Most mushroom-derived preparations and substances find use not as pharmaceuticals ("real" medicines) or botanical drugs, but rather as a novel class of products by a variety of names: dietary supplements (DS), tonics, functional foods, nutraceuticals, nutriceuticals, phytochemicals, mycochemicals, biochemopreventives, and designer foods.[14,40,41]

These terms vary in meaning from country to country, as does the regulation of these products. In the USA, the formal definition under the Dietary Supplements Health and Education Act (DSHEA) of 1994[42] exists only for DSs. "Functional foods" has no legal status or general acceptance, but is used as a definition in the dietetic profession. "Nutraceuticals" and "nutriceuticals" are also accepted definitions in the nutrition/science community,

TABLE 9. Proximate Composition of Fruit Bodies and Mycelia of *Coprinus comatus*[38]

	Content[a] (%)	
Component	Fruit bodies	Mycelia
Moisture	8.97 ± 0.08 A[b]	5.64 ± 0.75 B
Dry matter	91.03 ± 0.08 B	94.36 ± 0.75 A
Carbohydrate	58.37 ± 0.32 A	17.73 ± 0.58 B
Reducing sugar	15.91 ± 0.05 A	16.45 ± 0.22 A
Crude ash	8.44 ± 0.03 A	3.37 ± 0.43 B
Crude fat	3.11 ± 0.09 B	37.69 ± 1.00 A
Crude fiber	12.48 ± 0.16 A	14.44 ± 1.42 A
Crude protein	17.60 ± 0.12 B	26.77 ± 0.03 A

[a]Moisture and dry matter were presented based on air-dried mass; others were presented based on dry mass.
[b]Each value is expressed as mean ± standard deviation (n = 3). Means with different letters within a row are significantly different (P < 0.05).

TABLE 10. Content of Soluble Sugars and Polyols of Fruit Bodies and Mycelia of *Coprinus comatus*[38]

	Content (mg/g dry mass)	
Sugar or polyol	**Fruit bodies**	**Mycelia**
Glucose	1.43 ± 0.51 B[a]	2.37 ± 0.11 A
Myo-inositol	1.49 ± 1.07	nd[b]
Mannitol	78.76 ± 3.42 A	3.62 ± 0.08 B
Sucrose	nd	24.48 ± 0.31
Trehalose	169.14 ± 1.29 A	3.98 ± 0.39 B
Total	250.82 ± 4.23 A	34.45 ± 0.05 B

[a]Each value is expressed as mean ± SD (n = 3). Means with different letters within a row are significantly different (P < 0.05).
[b]nd: not detected.

but not embodied in law or regulation. "Medical foods" are regulated by the Food and Drug Administration (FDA) Office of Special Nutritionals, on a case-by-case basis, as enteral foods or foods for sick infants.[43,44]

DSs are ingredients obtained from foods, plants, and mushrooms (fungi) that are taken, without further modification, separately from foods for their presumed health-enhancing benefits.[45] The term DS includes one or more of the following dietary ingredients: (1) a vitamin, mineral, amino acid, herb, or other botanical; or (2) it is a dietary substance used to supplement the diet by increasing the total dietary intake and is intended for ingestion in the form of a capsule, powder, soft gel, or gel cap, and not represented as a conventional food or as a sole item of a meal or a diet (DSHEA, Public Law 103-417, 1994). Supplements are easy to add to the daily diet and are often the first steps consumers undertake towards greater nutritional

TABLE 11. Content of Free Amino Acids of Fruit Bodies and Mycelia of *Coprinus comatus*[38]

	Content (mg/g dry mass)	
Amino acid	**Fruit bodies**	**Mycelia**
L-Alanine	0.26 ± 0.01 A[c]	0.12 ± 0.01 B
L-Arginine	0.55 ± 0.01 B	1.33 ± 0.02 A
L-Aspartic acid	0.69 ± 0.07 A	0.09 ± 0.01 B
GABA[a]	0.63 ± 0.04 A	0.23 ± 0.04 B
L-Glutamic acid	2.54 ± 0.20 A	0.41 ± 0.02 B
Glycine	0.29 ± 0.01 B	0.32 ± 0.01 A
L-Histidine[b]	0.72 ± 0.03 A	0.41 ± 0.02 B
L-Isoleucine[b]	0.71 ± 0.11 A	0.20 ± 0.01 B
L-Leucine[b]	0.62 ± 0.01 A	0.35 ± 0.02 B
L-Lysine[b]	0.88 ± 0.02 A	0.16 ± 0.01 B
L-Methionine[b]	0.09 ± 0.01 B	0.30 ± 0.01 A
L-Phenylalanine[b]	0.80 ± 0.01 A	0.03 ± 0.01 B
L-Serine	0.41 ± 0.02 A	0.28 ± 0.02 B
L-Threonine[b]	nd[d]	0.23 ± 0.02
L-Tryptophan[b]	nd	0.72 ± 0.14
L-Tyrosine	0.55 ± 0.01 A	0.73 ± 0.14 A
L-Valine[b]	0.20 ± 0.01 A	0.14 ± 0.02 B
Total	9.94 ± 0.53 A	6.05 ± 0.41 B

[a]GABA, γ-aminobutyric acid. [b]Essential amino acid. [c]Each value is expressed as mean ± SD (n = 3). Means with different letters within a row are significantly different (P < 0.05). [d]nd: not detected.

TABLE 12. Proximate Composition of Fruiting Bodies of Studied Culinary-Medicinal Mushrooms[39]

Species/Fruiting body	Common name	Ash (%)	Carbohydrate (%)					Fat (%)	Protein (%)	Energy (kcal/ 100 g)	Ref.
			RS[a]	SP[a]	Fiber	DF[a]	Total				
Agaricus bisporus MS	Button mushroom	8.78	13.91	27.85	20.44	48.29	62.20	2.53	26.49	184.37	12
Agaricus brasiliensis (=*A. blazei*)	Brazilian mushroom	6.81	10.71	34.81	18.31	53.12	63.83	2.62	26.74	173.38	13
Agrocybe cylindracea brown	Willow-pine mushroom	7.90	14.73	28.85	22.80	51.65	66.38	3.54	22.18	179.50	14
Agrocybe cylindracea yellow	Willow-pine mushroom	6.65	7.88	45.83	19.54	65.37	73.25	3.63	16.47	130.07	13
Auricularia fuscosuccinea brown	Jin ear	4.02	9.87	61.32	11.69	73.01	82.88	4.48	8.62	114.28	15
Auricularia fuscosuccinea white	Snow ear	5.54	10.90	57.98	8.51	66.49	77.39	4.54	12.53	134.58	15
Auricularia mesenterica	Black ear	3.29	17.81	58.72	3.92	62.64	80.45	0.80	15.46	140.28	15
Auricularia polytricha	Red ear	2.05	17.62	70.52	3.63	74.15	91.77	0.48	5.70	97.60	15
Boletus edulis	King bolete	5.84	8.31	47.85	13.70	61.55	69.86	5.76	18.54	159.24	13
Clitocybe maxima cap	Big cup mushroom	5.28	22.21	21.44	26.78	48.22	70.43	2.70	21.59	199.50	16
Clitocybe maxima stipe	Big cup mushroom	3.67	31.47	19.45	31.76	51.21	82.68	1.24	12.41	186.68	16
Coprinus comatus	Shaggy ink cap	8.44	15.91	42.46	12.48	54.94	70.85	3.11	17.60	162.03	17
Cordyceps militaris	Northern *Cordyceps*	5.94	14.03	14.01	19.57	33.58	47.61	10.09	36.36	292.37	18
Trametes (=*Coriolus*) *versicolor*	Yun chih	6.37	28.27	36.82	23.24	60.06	88.33	1.10	4.20	139.78	19
Dictyophora indusiata	Basket stinkhorn	6.25	20.11	46.91	9.16	56.07	76.18	2.98	14.59	165.62	20
Flammulina velutipes white	Winter mushroom	6.93	13.88	34.31	15.99	50.30	64.18	8.89	20.00	215.53	21
Flammulina velutipes yellow	Winter mushroom	7.51	12.03	27.60	16.98	44.58	56.61	9.23	26.65	237.79	21
Ganoderma lucidum	Ling chih	1.77	15.12	10.90	59.16	70.06	85.18	5.13	7.92	138.33	19
Ganoderma lucidum antler	Ling chih	1.70	15.56	12.22	59.49	71.71	87.27	3.85	7.18	125.61	19

TABLE 12. Proximate Composition of Fruiting Bodies of Studied Culinary-Medicinal Mushrooms[39] (cont'd)

Species/Fruiting body	Common name	Ash (%)	Carbohydrate (%)					Fat (%)	Protein (%)	Energy (kcal/ 100 g)	Ref.
			RS[a]	SP[a]	Fiber	DF[a]	Total				
Ganoderma tsugae	Song shan ling chih	1.69	5.35	5.06	73.37	78.43	83.78	5.72	8.81	108.12	22
Ganoderma tsugae baby	Song shan ling chih	2.62	5.86	11.31	59.93	71.24	77.10	6.50	13.78	137.06	22
Grifola frondosa	Maitake	6.99	10.71	48.07	10.05	58.12	68.83	3.10	21.08	155.06	20
Hericium erinaceus	Lion's mane	9.35	17.39	39.63	7.81	47.44	64.83	3.52	22.30	190.44	20
Hypsizygus marmoreus	Hon-shimeji	7.75	15.69	24.78	28.09	52.87	68.56	4.09	19.60	177.97	23
Hypsizygus marmoreus white	Hon-shimeji	8.26	15.70	23.94	25.42	49.36	65.06	5.62	21.06	197.62	23
Inonotus obliquus	Chaga	8.36	4.92	48.27	34.58	82.85	87.77	2.36	1.51	46.96	24
Lentinus edodes 271	Shiitake	5.27	25.81	36.49	5.63	42.12	67.93	6.34	20.46	242.14	21
Lentinus edodes stipe	Shiitake	3.78	19.68	17.97	46.99	64.96	84.64	1.32	10.26	131.64	25
Lentinus edodes Tainung 1	Shiitake	5.85	21.91	41.99	4.88	46.87	68.78	5.71	19.66	217.67	21
Pleurotus citrinopileatus	Golden oyster	6.72	17.90	27.92	18.03	45.95	63.85	3.44	25.99	206.52	26
Pleurotus cystidiosus	Abalone mushroom	9.62	21.00	42.13	8.74	50.87	71.87	3.10	15.41	173.54	21
Pleurotus eryngii	King oyster mushroom	5.76	23.05	41.50	5.97	47.47	70.52	1.57	22.15	194.93	27
Pleurotus ferulae	Ferulae mushroom	6.60	30.46	30.67	12.98	43.65	74.11	3.01	16.28	214.05	16
Pleurotus ostreatus	Oyster mushroom	7.59	19.65	41.42	5.33	46.75	66.40	2.16	23.85	193.44	21
Pleurotus ostreatus gray	Oyster mushroom	5.90	23.21	40.99	9.82	50.81	74.02	3.76	16.32	191.96	16
Tremella fuciformis	Silver ear	6.14	31.64	50.08	2.91	52.99	84.63	0.93	8.30	168.13	15
Tricholoma giganteum	Giant mushroom	5.03	37.95	32.12	4.50	36.62	74.57	4.28	16.12	254.80	20

[a]RS, reducing sugar; SP, soluble polysaccharide; DF, dietary fiber = SP + fiber. [b]kcal/100 g = RS × 4 + fat × 9 + protein × 4.

TABLE 13. Proximate Composition of Mycelia of Studied Culinary-Medicinal Mushrooms[39]

Species/Mycelia	Common name	Ash (%)	Carbohydrate (%)					Fat (%)	Protein (%)	Energy[b] (kcal/100 g)	Ref.
			RS[a]	SP[a]	Fiber	DF[a]	Total				
Agaricus bisporus Tainung 3	Button mushroom	6.72	13.25	32.56	8.72	41.28	54.53	1.73	37.02	216.65	28
Agaricus brasiliensis	Brazilian mushroom	5.90	11.58	30.83	26.41	57.24	68.82	9.68	15.60	195.84	29
Agrocybe cylindracea brown	Willow-pine mushroom	5.12	17.48	13.17	28.96	42.13	59.61	17.32	17.95	297.60	14
Antrodia camphorata	Chang chih	3.92	25.39	31.27	20.14	51.41	76.80	9.79	9.49	227.63	29
Antrodia salmonea	Shiang shan chih	5.57	37.97	4.74	3.28	8.02	45.99	3.23	45.21	361.79	30
Coprinus comatus	Shaggy ink cap	5.26	36.45	12.33	20.38	32.71	69.16	2.96	22.62	262.92	17
Cordyceps militaris	Northern *Cordyceps*	6.97	12.97	24.57	19.17	43.74	56.71	7.20	29.12	233.16	29
Cordyceps sinensis	Caterpillar mushroom	3.59	29.26	21.80	10.18	31.98	61.24	11.76	23.41	316.52	30
Ganoderma tsugae	Song shan ling chih	5.18	15.42	11.56	19.39	30.95	46.37	21.86	26.59	364.78	22
Grifola frondosa	Maitake	3.95	10.19	23.11	11.70	34.81	45.00	24.65	26.40	368.21	31
Hericium erinaceus	Lion's mane	2.57	9.88	18.29	36.81	55.10	64.98	8.78	23.67	213.22	32
Hypsizygus marmoreus	Hon-shimeji	3.83	16.72	12.39	22.33	34.72	51.44	15.12	29.61	321.40	23
Hypsizygus marmoreus white	Hon-shimeji	3.69	19.10	6.40	22.40	28.80	47.90	20.62	27.79	373.14	23
Inonotus obliquus	Chaga	3.30	26.17	35.23	4.01	39.24	65.41	5.70	25.59	258.34	24
Morchella esculenta	Morel	5.03	12.21	22.72	6.30	29.02	41.23	12.03	41.71	323.95	31
Phellinus linteus	Sang hwang	1.91	19.04	35.02	11.57	46.59	65.63	4.04	28.42	226.20	33
Pleurotus citrinopileatus	Golden oyster	5.01	16.96	13.91	13.80	27.71	44.67	20.82	29.50	373.22	26
Sparassis crispa	Cauliflower mushroom	5.33	26.36	9.32	21.14	30.46	56.82	5.24	32.61	283.04	33
Termitomyces albuminosus	Termite mushroom	6.77	12.81	33.44	7.78	41.22	54.03	15.00	24.20	283.04	31

[a]RS, reducing sugar; SP, soluble polysaccharide; DF, dietary fiber = SP + fiber. [b]kcal/100 g = RS × 4 + fat × 9 + protein × 4.

awareness and the adoption of a healthy lifestyle. The DS pyramid has been developed recently, which, just like the food pyramid, has its own rules.

At the same time, the increase of interest in traditional remedies for various physiological disorders and the recognition of numerous biological activities of mushroom products has led to the coining of the term *mushroom nutriceuticals*,[14,25,32,36] not to be confused with nutraceuticals, functional foods, and pharmaceuticals. A mushroom nutriceutical is a refined, or partially refined, extract or dried biomass from either mycelium or the fruit body of a mushroom, which is consumed in the form of capsules or tablets as a DS (not a food) and has potentially therapeutic applications. Regular intake may enhance the immune response of the human body, thereby increasing resistance to disease and, in some cases, causing regression of the disease state. Thus, acting as immunopotentiators, mushroom preparations modify host biological responses[46–50] and are also known as biological response modifiers (BRMs).

The term DS ("nutriceutical") is broader than BRM and better describes the qualities of this class of substances. Another shortcoming of the term *biological response modifiers* in this context is that many of the mushroom preparations possess particular physiological effects, such as lowering of blood cholesterol or hepatoprotective activity.[25,36,37,40,41,46,50,51]

Several types of culinary-medicinal products are available on the market today:

Artificially cultivated fruit body powders, hot water or alcohol extracts of these, or the same extract concentrates and their mixtures.

Dried and pulverized preparations of the combined substrate, mycelium, and mushroom primordia after inoculation of edible semisolid medium (usually grains).

Biomass or extracts from mycelium harvested from submerged liquid culture grown in a fermentation tank.

Naturally growing, dried, mushroom fruit bodies in the form of capsules or tablets.

There is no doubt that medicinal mushroom–based products can serve as superior DSs. The problem is that mushroom-based DSs are so diverse, and there are currently no standard protocols for guaranteeing their product quality and critical testing. There is a serious need for improved quality and legal control, which are essential both to increase and maintain consumers' confidence, and to meet current and future standards set by the regulatory authorities.[25,32,37,40,41]

It should be mentioned that most suppliers and distributors of mushroom DSs provide very little and highly variable information on the source of their materials, ways of preparation, and composition. The field of mushroom DSs today is very far from unification and standardization.[50]

One of the difficulties facing this area is the lack of international consistency in the regulatory management of DSs *per se*. In some countries regulation is effectively based on a three-category system, i.e., foods, medicines, and DSs (or alternate term). In other countries, regulations pertaining to DSs (or other) generally sit under a broader (two-category) legislative system for either foods or medicines.

Recently, there has been growing popularity in developing MM products as DSs or functional foods. Significant questions arise with establishing DS and MM products, including their safety, standardization, regulation, efficacy, and mechanism of action.

A. Safety and Diversity of DS Types from Culinary-Medicinal Mushrooms

Safety of the substances considered is a central feature of any regulation measure. The substances might have obvious or hidden beneficial actions or exploit a placebo effect as long as the ideas of health are attached to them in the public's mind. However, their safety should be verified and proven as thoroughly as possible.

Drugs that affect body functions such as immune response, blood pressure, diuresis, and others are called pharmacodynamic agents. This is the way DSs are evaluated. Pharmacodynamics is the spectrum of biological responses produced by an intervention at a given time. The presence of therapeutic pharmacodynamic activity implies a lack of safety at a sufficiently high dose. This means that a mushroom preparation, like any other pharmacodynamic agent, cannot have pharmacological action without toxicological action. A completely safe agent would be without any activity whatsoever.[25,32,50–52]

It is commonly believed that many botanicals and mushrooms can be considered safe because of

their long history of use. The *Ganoderma lucidum* mushroom, for instance, was described in classical Chinese medicine (Shen Nong Ben Cao Jin, Compendium of Material Medica of the East Han dynasty) as long as 2000 years ago.[53]

However, safety to a pharmacologist is a relative concept, which is very different from the public notion of safety as an absolute concept. Here, we outline the reasons why we cannot take the safety of all mushroom-derived DSs for granted simply because they were used for many centuries in traditional human cultures.

Considering the historical perspective, "safety" in traditional terms is very different from that in modern times. First, mortality patterns of developed societies today are very different from those of traditional ones. Life expectancy in Japan today is 80 years or more, but in medieval China it was 20 years less (these figures do not reflect a maximum life span, but only a median age at death). Death itself (the likelihood of dying at a certain age) had a different kinetics in society. Second, traditional users rarely had the means to evaluate long-term or chronic toxicity of the agents. However, we do have cautionary instances of plants that have been used medicinally for centuries and recently proved to carry delayed toxic effects. An example of an ill effect is the mushroom *Paxillus involutus* (Batsch.:Fr.) Fr. It was widely used in Europe, and it is still taken as a food in many regions there. *Paxillus* syndrome was described only in 1971;[54] in fact, this syndrome is an immunohemolytic anemia. A patient who has eaten *Paxillus* over a long period (sometimes years) on occasion may develop *Paxillus* syndrome in a short time period of one or two hours after consumption. An antigen of still unknown structure stimulates a severe immune shock, resulting in such symptoms as diarrhea, subicterus, oliguria, anuria, hemoglobinuria, and renal pain.[50,54,55]

Many supposedly traditional mushroom products are now marketed in ways markedly different from those in the past. Today, larger amounts may typically be taken, or the material is used more frequently, or it is consumed in the form of enriched extracts, and it may be taken simultaneously with synthetic drugs. The user of Shiitake in old China, for instance, could not ingest as much active polysaccharide (Lentinan) as a modern user taking it in pure form extracted from Shiitake as a DS. Notably, 200 kg of fresh mushrooms are needed for extraction of 31 g of Lentinan.[56] This heightens the possibility of ill effects from traditionally "safe" mushrooms.

Also, many mushrooms or mushroom preparations traditionally taken as treatments for specific conditions are now often marketed for use as prophylactic agents. The idea of DSs, in many cases, implies that they are taken in the absence of any indicated conditions to prevent disturbances of health.

Finally, reliance on traditional use as an indication of safety involves a danger, namely, the poor information available to us from antiquity. Huxtable[52] recently carried out an analysis of historical sources of such information that are in many cases contradictory and vague. Information on acute toxicity may be located on the same page as the recommendations for use. The identities of the plants are sometimes dubious. Herbalists copied extensively from each other over thousands of years. The descriptions of symptoms are often vague and general (e.g., blood-stream disturbances or liver malfunction).[50]

Besides the problems of direct toxicity of some individual nutrients, nutrient supplementation can cause problems related to nutrient imbalances or adverse interactions with medications. Many problems associated with high doses of a single nutrient may reflect interactions resulting in a relative deficiency of another nutrient. Many examples are available from studies of DSs that have been used for a long time and are more common on the market then those derived from mushrooms. High amounts of calcium inhibit absorption of iron.[57] Folic acid can mask hematological signs of vitamin B_{12} deficiency, which, if untreated, can result in irreversible neurological damage. Folic acid can also interact adversely with anticonvulsant medications.[58] Abundant zinc supplementation can reduce copper status, impair immune responses, and decrease high-density lipoprotein cholesterol levels.

Because we lack many strict data on the safety of many mushroom preparations, we could only make estimates on the basis of available scientific knowledge. The clear advantages of using mushroom-based DSs with regard to safety (as opposed to herbal preparations) are the following:

The overwhelming majority of mushrooms used for production of DSs are cultivated commercially (and not gathered in the wild). This guaran-

tees proper identification and pure, unadulterated products. In many cases it also means genetic uniformity.

Mushrooms are easily propagated vegetatively and thus keep to one clone. The mycelium can be stored for a long time, and the genetic and biochemical consistency may be checked after considerable time.

The main advantage might be the fact that many mushrooms are capable of growing in the form of mycelial biomass in submerged cultures.[50,59]

Chang[48] suggests adopting the five "G" guidelines to enhance quality mushroom products. Although the text was originally oriented towards mushroom fruit bodies, it is still valid for other forms of mushroom products such as submerged culturing:

(1) GLP (Good Laboratory Practice): A known mushroom strain must be used; the source and nature of the strain culture must be clearly documented and should be properly maintained and preserved without contamination and degeneration.

(2) GAP (Good Agriculture Practice): Growth conditions must be known; the substrate should be free of heavy metals and composed of consistent levels of ingredients; the environmental conditions should include unpolluted air and a good sanitary growth area. The product should be harvested at optimal maturity and free of diseases.

(3) GMP (Good Manufacturing Practice): The parameters for processes must be known and maintained. The temperature, duration, and percentage of solvates used in extraction should be constantly monitored.

(4) GPP (Good Production Practice): The following tests must be conducted: a chemical analysis of the products to determine organic components and heavy metal contents; a microbial analysis to determine whether the type and level of micro-organisms present is within safe limits; and standardization of the formulation of the products.

(5) GCP (Good Clinical Practice): Medical practitioners must conduct high-quality clinical trials including double-blind studies, which should be conducted to ensure standardization and allow appropriate dosage determinations and product formulation for the effective treatment of a particular health problem.

The fifth G is somewhat problematic to DS developers. Once a clinical trial on humans is performed and bioactivity is confirmed, the product acquires a status of an unapproved active drug. Until the product is approved, it is banned from the market because it contains an unapproved drug. The process of drug approval is extremely expensive and lasts several years. By the time the product is approved, the manufacturers may loose their business. This problem requires that only large pharmaceutical companies run clinical trials, and those large companies do not engage in the research on traditional DSs. Another reason why DS developers do not wish to transform a healthy product into a medical product is because some health organizations, such as the FDA, do not approve a mixture of molecules, as in the case of mycelial or fruit body extracts, but only well defined, patentable molecules. On one hand, most DS marketers will be happy enough just to sell their mushroom products without printed health claims. They will advertise them in foreign markets that allow health claims on the Internet or by word of mouth. On the other hand, perhaps a whole new industry, or even industries, will arise having greater economic value than those currently producing mushrooms for food.

In 2004, the FDA published a guide for the industry: "Botanical Drug Products."[60] This guide explains when a botanical drug may be marketed under an over-the-counter (OTC) drug monograph and when FDA regulations require approval of a new drug application (NDA) for marketing. In addition, this document provides sponsors with guidance on submitting investigational new drug applications (INDs) for botanical drug products, including those currently lawfully marketed as foods (including conventional foods and dietary supplements) in the United States. Botanical products are finished, labeled products that contain vegetable matter as ingredients. A botanical product may be a food (including a dietary supplement) or a drug (including a biological drug). The term *botanicals* includes plant materials, algae, and macroscopic fungi. It does not include fermentation products such as extracts made from submerged fermentation of mushroom biomass. Also, a botanical drug substance does not include a highly purified or chemically modified substance derived from natural sources. The authors are not aware of any clinical trials in the US with mushrooms as the botanical drug.

B. Unsolved Problems of Culinary-Medicinal Mushroom-Based DSs

Unfortunately, standardization around the world of DSs from MMs is still in its early stages, including an insufficient understanding of DS bioactive effects. We don't have internationally recognized standards and protocols for production and the testing of MMs products. Only proper standards and protocols can guarantee product quality. Without consistency in the quality of MM products, commercially available MM preparations of mushrooms will be dramatically different and vary enormously in composition and effectivity. It is not known whether bioactive effects are caused by a single component or are the result of a synergistic impact from several ingredients. There are insufficient data to determine which components work better, mushroom fruiting bodies or submerged mycelia powder vs. extracts. Are simple dried fruiting bodies and mycelium powders as effective as hot water, alcoholic, or hydro-alcoholic extracts? Between crude extracts and isolated fractions, which one is more effective and has a higher safety profile (some companies are selling fraction-like *Grifola frondosa* maitake D-fraction or GLPS fraction of *Ganoderma lucidum*)? The role of low-molecular-weight compounds for MM extracts is still unclear.[14,25,32,50]

What is more effective—the combination of components containing biomass or extracts of 2–10 different species in one pill, or having one species in one pill? How can one assess the effectiveness of different mushroom products when blending in many species in one product ("shotgun" approach)? Since mushroom products can be cytokine stimulants, from what age are they safe for young children to take, since their immune systems are not yet mature? What dosages are safe and effective during pregnancy and nursing? The absence of sufficiently elaborated standards for the recommended use of MM DSs, including precise doses and duration of administration, needs very serious investigation. Some research shows that too high a dose could lead to immune suppression; too low a dose might not trigger an immune response. Furthermore, major problems associated with mushroom-based DSs are due to their wide variability, the current lack of standards for production, and the lack of testing protocols necessary to guarantee product quality. The active ingredients of many present-day commercial mushroom products have not been indicated.[14,50]

Adulteration of MM products with spurious species (for instance, different *Ganoderma* species for *Ganoderma lucidum*, *Stereum* species for *Trametes* species, and different *Cordyceps* species for *C. sinensis*) is very common. Difficulties are found in producing pure β-glucans for the market (90%–95% of β-glucan on the market is considered counterfeit and adulterated). Adulteration has led to a number of adverse effects, resulting in nephropathy, acute hepatitis, coma, and fever. There have also been neurological, cardiovascular, and gastrointestinal problems reported with TCM.[61–62]

We still have not solved the problems concerning the safety of several well-known MM products. For example, there have been several reports from hospitals in Japan where the intake of MM *Agaricus brasiliensis* products caused liver dysfunctions.[63] On the basis of a study of a phase I/II trial of polysaccharide extract of *Grifola frondosa* in breast cancer patients, it was concluded that the maitake mushroom produces more complex effects than presumed and may depress as well as enhance immune function.[64]

What is the role of fresh mushroom consumption? The consumption of fresh mushrooms was found to increase, for example, anti-β-glucan antibodies in the serum of humans. It was also suggested to provide better defense against pathogenic fungi by the Ohno group from Japan.[65]

Information is still lacking on the use of MM DSs and their intercrossing or interaction with some drugs.[14]

VII. QUALITY CONTROL AND REGULATIONS OF MEDICINAL MUSHROOM DSs AND MEDICINAL MUSHROOM PRODUCTS

A. World Health Organization Guidelines

In 1991, the World Health Organization (WHO) published its "Guidelines for the Assessment of Herbal Medicines."[66] This document was created over a period of five years (1986–91) at several International Conferences of Drug Regulatory Authorities and is based on the WHO's recognition that 80% of the world's population in developing countries rely on traditional medicine; "a major part of the traditional therapies involves the use of active constituents of plant extracts"; and "consid-

erable growth has occurred in popular, official, and commercial interest in the use of natural products."

The objective of the guidelines "is to define basic criteria for the evaluation of quality, safety, and efficacy" of all herbal (as usual, including mushrooms among the herbs) medicines. "As a general rule in this assessment, traditional experience means that long-term use as well as the medical, historical, and ethnological background of those products shall be taken into account."[66] Depending on each country's situation, "the definition of long-term use may vary, but would be at least several decades. . . . Prolonged and apparently uneventful use of a substance usually offers testimony of its safety."

According to Alkerele,[67] "safety should be overriding criterion in the selection of herbal medicines for use in health service systems." The WHO guidelines call for various assessments of quality, safety, efficacy, and intended use of herbal medicines. The guidelines call for reference to pharmacopoeia monographs, if they exist. If none exist, a manufacturer applying for marketing licenses or registration should supply a monograph with the same components as an official pharmacopoeia. All procedures should correspond to GMP, and there needs to be standard testing of the product in its final packaging form. The assessment should also make a distinction between old and new combination products.[14,40,50]

With regard to safety, "a guiding principle should be that if the product has been traditionally used without demonstrated harm, no specific restrictive regulatory action should be undertaken unless new evidence demands a revised risk-benefit assessment."[67]

Finally, consumer product information is recommended, including a quantitative list of active ingredients, dosage, dosage form, indications, mode of administration, duration of use, any major adverse effects, contraindications, warnings, etc.[14,25,40,50]

B. Codex Alimentarius

The Codex Alimentarius Commission implements the Joint Food and Agriculture Organization (FAO)[68] and World Health Organization (WHO) Food Standards Programme, the purpose of which is to protect the health of consumers and to ensure fair practices in the food trade. The Codex Alimentarius (Latin, meaning food law or code) is a collection of internationally adopted food standards presented in a uniform manner. It also includes provisions of an advisory nature in the form of codes of practice, guidelines, and other recommended measures to assist in achieving the purposes of the Codex Alimentarius. The commission has expressed the view that codes of practice might provide useful checklists of requirements for national food control or enforcement authorities. The publication of the Codex Alimentarius is intended to guide and promote the elaboration and establishment of definitions and requirements for foods, to assist in their harmonization, and, in doing so, to facilitate international trade.[68]

The 28th Session of the Codex Alimentarius Commission was held during July 2005. This event has been the subject of considerable controversy, in part because many member countries regulate substances as therapeutic goods or pharmaceuticals and not as foods (if they were not foods, they would be excluded from the Codex Alimentarius). The Codex does not seek to ban supplements, but to subject them to labeling and composition requirements. Some groups have pointed to greater concerns related to restrictions on dietary supplement ingredients in Europe via the European Food Supplements Directive (which utilizes approved lists of ingredients and ingredient forms) and potentially restrictive dosage limits to be based on a Codex model via the FAO and WHO Nutrient Risk Assessment Project. The Codex regulations have been rejected by the FDA in the US but accepted by the EU Trading Bloc.[40]

Below we describe accepted rules for quality control and regulations of medicinal mushroom DSs and medicinal mushroom products in the USA, EU, Australia and New Zealand, Japan, and P.R. China.

1. The United States

In the United States, virtually all facets of DS manufacturing, labeling, and advertising are covered by extensive regulations issued and enforced by the FDA[69] and by the Federal Trade Commission (FTC).[70]

For decades the FDA regulated DSs as foods to ensure that they were safe and wholesome and that their labeling was truthful and not misleading. An important facet of ensuring safety was the FDA's evaluation of the safety of all new ingredients, in-

cluding those used in DSs, under the 1958 Food Additive Amendments to the Federal Food, Drug, and Cosmetic Act.[71] "GRAS" is an acronym for the phrase Generally Recognized as Safe. Under sections 201(s) and 409 of the Federal Food, Drug, and Cosmetic Act, any substance that is intentionally added to food is a food additive, which is subject to pre-market review and approval by FDA, unless the substance is generally recognized, among qualified experts, as having been adequately shown to be safe under the conditions of its intended use, or unless the use of the substance is otherwise excluded from the definition of a food additive. For example, substances whose use meets the definition of a pesticide, a dietary ingredient of a dietary supplement, a color additive, a new animal drug, or a substance approved for such use prior to September 6, 1958, are excluded from the definition of food additive. Sections 201(s) and 409 were enacted in 1958 as part of the Food Additives Amendment to the Act. While it is impracticable to list all ingredients whose use is generally recognized as safe, FDA published a partial list of food ingredients whose use is generally recognized as safe to aid the industry's understanding of what did not require approval. The use of a food substance may be GRAS either through scientific procedures or, for a substance used in food before 1958, through experience based on common use in food. Interestingly, 4-hydroxymethyl-benzenediazonium (HMBD) ion has been detected in the commonly cultivated and consumed mushroom *Agaricus bisporus* at levels near 0.6 ppm.[72] HMBD is a fungal metabolite of agaritine, a naturally occurring substance found in *Agaricus* species. HMBD has been implicated in causing DNA strand breaks at 1.2 mg per 70-kg person.[73] The calculated safe dose is less than 4 g mushroom per day or 1 meal every 100 days. However, since mushrooms are GRAS, HMBD is exempt in this food.

However, with passage of the DSHEA, Congress amended the Food, Drug, and Cosmetic Act to include several provisions that apply only to DSs and dietary ingredients of DSs. As a result of these provisions, dietary ingredients used in DSs are no longer subject to the pre-market safety evaluations required for other new food ingredients or for new uses of old food ingredients. However, they must satisfy several other safety provisions. The Council for Responsible Nutrition (CRN) outlines different regulations as they apply to foods, DSs, and drugs when reviewing products.[74]

The DSHEA acknowledges that millions of consumers believe that DSs may help to augment daily diets and provide health benefits. Congress's intent in enacting the DSHEA was to meet the concerns of consumers and manufacturers to help ensure that safe and appropriately labeled products are available to those who want to use them. In the findings associated with the DSHEA, Congress stated that there may be a positive relationship between sound dietary practice and good health, and that, although further scientific research is needed, there may be a connection between dietary supplement use, reduced healthcare expenses, and disease prevention.

Today, DS manufacturers that wish to market a new ingredient (i.e., an ingredient not marketed in the United States before 1994) have two options. The first involves submitting to the FDA, at least 75 days before the product is expected to go on the market, information that supports their conclusion that the new ingredient can reasonably be expected to be safe. "Safe" means that the new ingredient does not present a significant or unreasonable risk of illness or injury under the conditions of use recommended in the product's labeling. This information submitted by the manufacturer becomes publicly available 90 days after the FDA receives it. CRN believes that newly isolated polysaccharide from mushrooms would not require pre-market 75-day notification.[50,75]

Another option for manufacturers is to petition the FDA to establish the conditions under which the new dietary ingredient may reasonably be expected to be safe. To date, FDA's Center for Food Safety and Applied Nutrition has received no such petitions.

Furthermore, makers of natural supplements must derive their products from substances that are considered safe. If a manufacturer wants to market a product containing an ingredient that was not used in commerce prior to the passage of the DSHEA, the FDA must be notified in advance and provided with safety data. At any time, the FDA is fully authorized to require manufacturers to recall a product if the agency believes it presents a health risk.

Under the DSHEA, makers of DSs are also limited as to the type of claims they can make

about their products. "Statements of nutritional support," commonly referred to as "structure/function" (SF) claims, are restricted to explaining how a particular product or ingredient affects the structure or function of the body. A supplement maker may not claim that a product is intended to "diagnose, treat, cure, or prevent a disease." The manufacturer must have substantiation that the statement is truthful and not misleading and must notify the FDA within 30 days after first using the claim in the marketplace.

One of the most important matters of regulation, in which the FDA has invested much effort in recent years, is the labeling of DS products. The new nutrition-labeling format for DSs includes a supplement facts box, which lists vitamins and minerals always in the same order. The new labels show the amount of each nutrient in terms of metric quantities (e.g., 60 mg of vitamin C) and in terms of percentages of the DV, if one has been established. All ingredients in the product must be listed, either within the fact box or in a separate list of ingredients below the fact box. For extracts, additional information may be provided including the solvent used and the concentration of the extract.[40,50]

The final rules for SF claims for DSs under DSHEA. In April 1988, the FDA published proposed regulations that included a highly controversial redefinition of the word *disease*. Many industry and consumer groups viewed this as an attempt to restrict the scope of the SF claims under DSHEA. Since then, the FDA has received 235,000 comments from the public about this measure: 213,000 as form letters circulated by consumer and trade groups and 22,000 as individual letters from consumers, members of industry, and other interested parties.[40,50,75]

Formerly, the definition of disease was "any deviation from, impairment of, or interruption of the normal structure of function. . . ." Now, the FDA will use the definition "damage to an organ, structure, or system. . . ." By this alternative definition, the FDA automatically reduces the range or number of health claims for DSs.

On January 6, 2000, the FDA issued its final regulations on SF claims for DSs under the DSHEA of 1994 (65 FR 1000-1050). In these rules, a significant shift is made in relation to SF claims for over-the-counter drugs, including DSs. In particular, the FDA has enlarged the range of SF claims by agreeing with the comments by the American Herbal Products Association (AHPA) that some claims are not disease claims, but instead are claims that deal with the structure or function of the body. Thus, DSs will be able to make claims for antacid, digestive aid, short-term laxative, and other uses previously prohibited to DSs.[40,69,75]

Many examples of conditions and statements for which SF claims are allowed under the DSHEA are worked out by these regulations. They include many that are important for mushroom-derived DSs, such as "immune system function," "maintenance of cholesterol levels that are already within the normal range," "tonic," and "as part of the diet to maintain blood sugar levels." Many other examples remain disease claims, such as "anti-inflammatory," "lowers cholesterol," and "controls blood sugar in persons with insufficient insulin." Another important regulation states that "the name of a product should not contain the name or recognizable portion of the name of the disease."[40,50,75]

Thus, the FDA regulates all key steps of DS development and application in the USA. The seemingly important drawback of the FDA empowerment is that the consumers have to be injured prior to banning the product on the shelf. However, in case the FDA has indications of a DS to be unsafe, it can issue a warning that might have an impact on the consumer.

In April 2005, the CFSAN/Office of Nutritional Products, Labeling, and Dietary Supplements released the Dietary Supplement Labelling Guide. The guide applies to both dietary supplement ingredients and finished dietary supplement products manufactured or produced in the US as well as those produced in other countries. The guide covers all aspects of dietary supplement labeling, including requirements related to statements of identity, net quantity listing supplement facts panels, ingredient listing, health and structure/function claims, and pre-market notifications for new dietary ingredients. There are several types of claims:[76,77]

(A) An authorized health claim is an "explicit or implied characterization of a relationship between a substance and a disease or a health-related condition." According to the FDA, a health claim "describes the effect a substance has on reducing the risk of or preventing a disease, e.g., 'calcium may reduce the risk of osteoporosis.'" This type of

claim requires significant scientific agreement and must be authorized by the FDA.

(B) A qualified health claim is supported by less scientific evidence than an authorized health claim. The FDA requires that qualified claims be accompanied by a disclaimer that explains the level of the scientific evidence supporting the relationship.

(C) A structure/function claim describes the role of a substance intended to maintain the structure or function of the body. Structure/function claims do not require pre-approval by the FDA. Structure/function claims are permitted if such a statement is truthful and not misleading, includes a required disclaimer, and the marketer notifies the FDA of the claim no later than 30 days after the first marketing of the product.

When the FDA learns of a supplement claim it believes violates the provisions of the Food, Drug and Cosmetic Act or regulations promulgated thereunder, it sends a letter to the supplement manufacturer or marketer advising them of the questionable claims, the potential violations raised by those claims, and a request that the claims be withdrawn. The FDA posts these letters in a database on the Internet that is searchable by date, month, and type of violation. Responses to these letters are also posted when received.

The guide is intended to address questions about dietary supplement labeling issues and reflects the FDA's current position on these issues. However, it is not legally binding or precedent-setting but rather is merely an interpretation of the applicable statutes and regulations to educate the industry on how the FDA believes compliance can be met.[40]

Several examples of the actions the FDA took against violators of the health claims regulations are available. In December 2001, FDA's New York District Office recommended detention without physical examination for Essence of Mushrooms capsules, 400 mg. The product, manufactured by Windsor Health Products Ltd., Kowloon, Hong Kong, was shipped as vitamins via Federal Express. However, the FDA examination found accompanying labeling promoting the product for treatment of cancer. In addition, the labeling also identified the manufacturer's Web site, which was found to be promoting the Essence of Mushrooms as an alternative therapy for cancer. FDA refused entry of the product.

In October 2006 the FDA announced a public hearing on the regulation of certain conventional foods that companies are marketing as "functional foods." The purpose of the hearing was for the agency to share its current regulatory framework and rationale regarding the safety evaluation and labeling of these foods, and to solicit information and comments from interested persons on how the FDA should regulate these foods under the agency's existing legal authority.[40]

In December 2006, the Senate passed the Dietary Supplement and Nonprescription Drug Consumer Protection Act, dubbed "the AER bill" (S.3546.ES). The legislation will amend the federal Food, Drug and Cosmetic Act to require the reporting of "serious" adverse events for both over-the-counter (OTC) drugs and dietary supplements to the US FDA. Under this bill, companies would be required to include contact information on their products' labels for consumers to use in reporting adverse events. They would further be required to notify the FDA of any serious adverse event reports received within 15 business days. According to the bill, the term *adverse event* means any health-related event associated with the use of a nonprescription drug that is adverse, including:

(A) An event occurring from an overdose of the drug, whether accidental or intentional;

(B) An event occurring from abuse of the drug;

(C) An event occurring from withdrawal from the drug; and

(D) Any failure of expected pharmacological action of the drug.

The term *serious adverse event* is an adverse event that results in death, a life-threatening experience, in-patient hospitalization, a persistent or significant disability or incapacity, or a congenital anomaly or birth defect, or requires, based on reasonable medical judgment, a medical or surgical intervention to prevent one of the above outcomes. Among these key provisions are those requiring that only serious adverse events, not just any complaint, be reported; exempting retail stores from reporting directly to the FDA; and preempting a potential patchwork of state laws on the issue.[40,50]

2. The European Union

In Europe, the DSs are known mostly under the name of "food supplements." "Health food" is a

marketing term rather than a legal term, normally denoting a product sold in specialty stores.[40,43] In addition, the term DS is used by the European Union for products regulated as Foods for Particular Nutritional Uses (PARNUTS).

Consumer safety is Europe's primary concern. A good example of the statement is that individual countries have already banned several supplements available in the USA. European countries follow the precautionary principle, which means that when they suspect a product may cause harm, they do not wait for proof before they take action against it. For example, the FDA warned American consumers in March 2002 that supplements containing kava, promoted as a relaxant, may cause liver injury—sometimes so severe that a transplant is required, sometimes even fatal, based on the case of a 45-year-old woman using kava, who suddenly required a liver transplant, as well as reports of 25 similar cases in Europe. However, while the FDA issued only a warning, several European countries, including Germany and France, banned the sale of kava.[78]

As the European countries vary in the ways they regulate DSs, the European Commission (EC) aims to provide a uniform standard of quality and safety for all member states. The EC divided DSs into two groups for the purpose of regulation: vitamins/minerals and herbals. The groups are treated separately.

Directive 2002/46/EC, which harmonized EC legislation on vitamins and minerals as food supplements, was published in the Official Journal of the EC (L183/51) on June 10, 2002 (Directive 2002/46/EC). The directive defined the term *food supplements* and set out labeling requirements. The directive did not immediately outlaw any products already on the market.[79]

The set standards enable a product approved in one country to be sold in other member countries. Also, everything in the EC directive follows from the basic principle that, under normal circumstances, a balanced diet can provide the necessary nutrients for development and health. For instance, the directive forbids manufacturers to state or even suggest the contrary; a manufacturer may not state that a supplement is a substitute for a varied diet, or that it can prevent, treat, or cure a disease. Furthermore, rather than setting only general guidelines, the directive lists exactly which vitamins and minerals may be sold in member countries, and in what form.[78]

Despite the directive's apparent advantages for the consumer, there are some disadvantages. It will tighten regulation in some countries, but loosen it in others that previously had stricter laws. It also allows each country to decide whether it will require government notification from manufacturers when a new supplement comes on the market.

The EC is developing a list of alternative herbal substances. It includes details about each substance: what it is to be used for, the strength and daily dose, how it is to be administered, and possible drug interactions or adverse effects. The date tentatively set for EU countries to pass appropriate laws regarding herbal medicines was December 31, 2004.[40,80]

The EU relates to herbal DSs in the same way it does to conventional medications. European countries vary in the degree to which they use herbals and how carefully they regulate them. Germany, Austria, Switzerland, and France lean toward a stricter approach, while the Netherlands and the United Kingdom have traditionally been more lenient. The same products sold as medicine in Germany are sold as food supplements in the Netherlands.[40,78]

Germany was in the vanguard of the move toward stricter herbal regulation for all EU countries, resulting in Directive 2001/83/EC. Although the directive does not relate to homeopathic medicines, it does cover all herbals used in the treatment or prevention of disease or as modifiers of physiological functions, including inducing relaxation. Therefore, under the directive, "natural" will no longer automatically mean "good." Manufacturers are not allowed to claim that a product is safe and effective simply because it is natural.[40,78] EC legislation does not refer to mushroom products directly.

The dietary supplements industry in Europe strongly opposed the Food Supplements Directive (2002/46/EC). A large number of consumers throughout Europe, including over one million in the UK, and many doctors and scientists, have signed petitions against what are viewed by the petitioners as unjustified restrictions of consumer choice. The European Court of Justice ruled on July 12, 2005, that the directive is valid, although the court's own advocate general advised that the

declaration was invalid under EU law.[81]

At the end of 2006, Regulation (EC) No 1924/2006 on the use of nutrition and health claims for foods was adopted by the Council and Parliament. This regulation made standard rules for the use of health or nutritional claims (such as "low fat," "high fibre," and "helps lower cholesterol") on foodstuffs based on nutrient profiles. The Health Claims Regulation ensures that any claim made on food labels in the EU is clear, accurate, and substantiated. In doing so, it enables consumers to make informed and meaningful choices when it comes to food and drinks. This also contributes to a higher level of human health protection, as it ties in with the EC campaign for healthier lifestyle choices by allowing citizens to know exactly what they are consuming. The regulation also aims to ensure fair competition and promote and protect innovation in the area of food. Only products offering genuine health or nutritional benefits will be allowed to use these terms on their labels.[82]

3. Australia and New Zealand

The use of natural products in Australia has been widespread for many years, as evidenced by the presence of an active organization of herbalists formed in 1920.[83] In 1985, the Australian Parliament established the Working Party on Natural and Nutrition Supplements to review the quality, safety, efficacy, and labeling of herbs and related products, including mushrooms for appropriate regulation under the Therapeutic Goods Act of 1990.[40,50]

The Working Party's report dealt with approximately 1,144 herbs and "therapeutic substances," dividing them into three groups, each having a recommended level of control and labeling, "appropriate to their level of toxicity".

Just like in the case of the EU and the US, food regulations between New Zealand and Australia were awaiting harmonization and standardization. The basic body responsible for this was the Australia New Zealand Food Authority (ANZFA). ANZFA's role is to protect the health and safety of people in Australia and New Zealand by maintaining a safe food supply. ANZFA is a partnership between the Commonwealth Government, Australian States and Territories governments, and the New Zealand Government. As an independent expert body, ANZFA is responsible for developing and reviewing food standards for both Australia and New Zealand. ANZFA makes recommendations to change the food standards to the Australia New Zealand Food Standards Council, a Ministerial Council made up of Commonwealth, State and Territory, and New Zealand Health Ministers. If the council approves the recommendations made by ANZFA, the food standards are automatically adopted as regulations into the food laws of the Australian States and Territories and New Zealand.[40,84]

Specific regulations were applied to novel and special-purpose foods. DSs derived from fungi are included (along with herbs and species) into the category of novel foods. "Novel food" means a non-traditional food (a food that does not have a history of significant human consumption by the broad community in Australia or New Zealand) for which there is insufficient knowledge in the broad community to enable safe use in the form or context in which it is presented, taking into account:

(A) The composition or structure of the product;

(B) Levels of undesirable substances in the product;

(C) Known potential for adverse effects in humans;

(D) Traditional preparation and cooking methods; or

(E) Patterns and levels of consumption of the product.

Traditional foods of a particular community may be considered novel if they are made available to a new or wider community without adequate information regarding applicable presentation and use. Novel foods can take on many forms, and not all are natural substances. There is an increase in the number and variety of novel foods on the market as a result of technological developments, trade opportunities, scientific advances, and increasing ethnic diversity of the population.[40,84]

The New Zealand Food Safety Authority (NZFSA) was established in July 2002 to improve the effectiveness of New Zealand's food safety system by coordinating and harmonizing food safety efforts. Specifically, New Zealand wanted to address inconsistencies between the methods used in the Ministry of Agriculture and Forestry's export food safety program and the Ministry of Health's domestic food safety program. NZFSA has farm-to-table responsibilities—from primary production through processing to retailers, importing, and

exporting, as well as responsibility for consumer education.[85] NZFSA has two main areas of focus: to protect and promote public health and safety through the administration of food-related legislation and to facilitate access to markets for NZ food products and related products.[86]

4. Japan

In Japan, Food with Health Claims (FHC) refers to foods that comply with the specifications and standards established by the Minister of Health, Labor and Welfare (MHLW) and are labeled with certain nutritional or health functions. These foods are categorized into two groups, according to differences in purpose and function: (a) Foods for Specified Health Uses (FOSHU): foods officially approved to claim their physiological effects on the human body; and (b) Foods with Nutrient Function Claims (FNFC): foods that are labeled with the functions of nutritional ingredients (vitamins and minerals).[40]

Under Japan's Nutrition Improvement Act of 1992 the regulatory system of FOSHU was developed to delineate "functional foods" from the pharmaceutical regulations. The Japanese government defines FOSHU as "foods that are expected to have certain health benefits, and have been licensed to bear a label claiming that a person using them for a specified health use may expect to obtain the health use through the consumption thereof."[40,87] The classification or list has no status outside Japan. As of January 8, 2004, FOSHU foods totaled 402 items. Among them, 91 products have been approved since the beginning of 2003. As of 2005, FOSHU foods account for US$6 billion with over 500 approved products. The total market value including non-FOSHU functional foods and DSs is three times higher. FOSHU refers to foods containing ingredients with functions for health and officially approved to claim its physiological effects on the human body. FOSHU is intended to be consumed for the maintenance/promotion of health or special health uses by people who wish to control health conditions, including blood pressure or blood cholesterol. In order to sell a food as FOSHU, the assessment for the safety of the food and effectiveness of the functions for health is required, and the claim must be approved by the MHLW.

Requirements for FOSHU approval include:

(A) Effectiveness on the human body is clearly proven;

(B) Absence of any safety issues (animal toxicity tests, confirmation of effects in the cases of excess intake, etc.);

(C) Use of nutritionally appropriate ingredients (e.g., no excessive use of salt, etc.);

(D) Guarantee of compatibility with product specifications by the time of consumption;

(E) Established quality control methods, such as specifications of products and ingredients, processes, and methods of analysis.

In addition to "regular" FOSHU, new types of FOSHU were introduced to facilitate applicants for FOSHU approvals:

(A) Qualified FOSHU: Food with health function, which is not substantiated on scientific evidence that meets the level of FOSHU, or the food with certain effectiveness but without established mechanism of the effective element for the function, will be approved as qualified FOSHU.

(B) Standardized FOSHU: Standards and specifications are established for foods with sufficient FOSHU approvals and accumulation of scientific evidence. Standardized FOSHU are approved when it meets the standards and specifications.

(C) Reduction of disease risk FOSHU: This claim is permitted when reduction of disease risk is clinically and nutritionally established in an ingredient. FOSHU regulated products are typically novel products and not value-added general-purpose foods. The majority of products are specifically focused on prevention of disease and maintenance of health status rather than the direct treatment of disease states. Health agencies of the Japanese government develop and maintain the regulatory framework that covers products that are to be regulated under the FOSHU system.[87]

FOSDU refer to foods that are approved/permitted to display that the food is appropriate for specified dietary use. There are five categories of FOSDU:

(1) Formulas for pregnant or lactating women;

(2) Infant Formulas;

(3) Foods for the elderly with difficulty in masticating or swallowing;

(4) Medical foods for the ill; and

(5) FOSHU.

Prohibition of Exaggerated and Misleading Claims (under Health Promotion Law): Any

claims related to health or function made on functional foods must be relevant and substantiated by scientific evidence. Advertisements are used in different media, to promote food sales. Some are advertised to convey positive effects on health maintenance and promotion, not necessarily having scientific evidence on the claim. When these advertisements are not regulated and uncontrolled, consumers who believed the claim might miss an opportunity for an adequate medical consultation, and therefore experience adverse effects on health. Under Paragraph 2, Article 32, of the Food Promotion Law, exaggerated and misleading claims are prohibited.[40,87]

5. P. R. China

Health foods or DSs are defined to be foodstuffs that have specific health functions or that provide an additional source of vitamins and minerals. They are recommended to be used for specific groups of people for the purposes of adjusting their physical functions and to enhance their immune responses. Use of health foods or DSs is not recommended for the curing of diseases, and also should not have any short- or long-term harmful effects to the human body.

In 1996, health food regulations were issued by the Ministry of Health, comprising seven chapters and 35 articles [http://baike.baidu.com/view/436952.htm]. A revised version was published on 30 April 2005 by the State Food and Drug Administration (SFDA) [http://www.sda.gov.cn/WS01/CL0053/24516.html]. The updated regulations contain eight chapters and 105 articles. Four new annexes were also added.

The regulations cover a wide range of quality and safety control issues relating to health foods/DSs in P. R. China, including application and approval procedures; production and management; labeling and marketing; and supervision and penalties.[4,5,25]

VIII. MUSHROOMS AS A MEDICINE

Today, a total of 126 medicinal functions are thought to be produced by medicinal mushrooms and fungi, including antitumor, immunomodulating, antioxidant, radical scavenging, cardiovascular, anti-hypercholesterolemia, antiviral, antibacterial, anti-parasitic, antifungal, detoxification, hepato-protective, and anti-diabetic effects.[14,88-89]

According to current categories of botanical products, MM can serve as botanical drugs or real drugs. Botanical drugs products are complex extracts to be used for treatment of disease. The guidelines for this relatively new regulatory category were released in 2004 (http://www.fda.gov/cder/Guidance/4592fnl.htm).[60] Botanical drugs are clinically evaluated for safety and efficacy just like conventional drugs, but the process for botanical drugs can be expedited because of the history of safe human use. Botanical drugs are highly but not completely characterized and are produced under the same strictly regulated conditions as conventional pharmaceuticals.[90–91]

Drugs can be prescription drugs or over-the-counter drugs. These products require the most rigorous testing, including three distinct phases of clinical testing to ensure safety and efficacy, and close scrutiny by the FDA. Although most early pharmaceutical products were botanical preparations and at least 25% of the pharmaceuticals used today are based on plant-derived products,[92] only pure compounds isolated from plants and subjected to the same rigors as synthetic pharmaceuticals can be conventional drugs.[93]

Mushroom compounds, especially polysaccharides, prevent oncogenesis, show direct antitumor activity against various synergetic tumors, and prevent tumor metastasis. Their activity is especially beneficial when used in conjunction with chemotherapy. The antitumor action of polysaccharides requires an intact T-cell component; their activity is mediated through a thymus-dependent immune mechanism. They activate cytotoxic macrophages, monocytes, neutrophiles, natural killer cells, dendritic cells, and chemical messengers (cytokines, such as interleukins, interferons, and colony-stimulating factors) that trigger complementary and acute phase responses. Also, mushroom polysaccharides can be considered as multicytokine inducers able to induce gene expression of various immunomodulatory cytokines and cytokine receptors.[46,47,94–97]

The benefits of mushroom compounds on different clinical conditions have attracted the interest of the scientific community in the last decade in order to understand the molecular mechanisms responsible for their actions. Several classes of mushroom compounds, such as proteins, polysaccharides, lipopolysaccharides, and glucoproteins,

have been classified as molecules that have potent effects on the immune system. They may restore and augment immunological responses of host immune effector cells, but they have no direct cytotoxic effect on tumors.[98]

Polysaccharides from other less known but promising mushroom species also show positive results in treating cancers *in vitro* and *in vivo*. These species include *Agaricus brasiliensis*, *Phellinus linteus*, *Grifola frondosa*, *Tremella mesenterica*, *Hypsizygus marmoreus*, *Flammulina velutipes*, and others. A new class of antitumor MM drugs has been called biological response modifiers (BRMs). The application of BRMs has become the new kind of cancer treatment, together with surgery, chemotherapy, and radiotherapy.[14,46,47,98]

Immunoceuticals isolated from more than 30 MM species have demonstrated antitumor activity in animal treatments. However, only a few have been tested for their anticancer potential in humans. The few that have been tested are β-D-glucans or β-D-glucans linked to proteins. Moreover, the latter have shown greater immunopotentiation activity than the free glucans. There are plenty of clinical studies proving the cancer inhibitory effects of *Lentinus edodes*, *Grifola frondosa*, *Schizophyllum commune*, *Ganoderma lucidum*, *Trametes versicolor*, *Inonotus obliquus*, *Phellinus linteus*, *Flammulina velutipes*, *Codyceps sinensis*, etc. Mushroom immunoceuticals act mainly by elevating the host immune system. This process includes activation of dendritic cells, NK cells, T-cells, and macrophages, and production of cytokines. Several MM products, mainly polysaccharides and especially β-glucans, were developed with clinical and commercial purposes: Krestin (PSK) and PSP (polysaccharide peptide) from *Trametes versicolor*; Lentinan, isolated from *Lentinus edodes*; Schizopyllan (Sonifilan, Sizofiran, or SPG), from *Schizophyllum commune*; Befungin from *Inonotus obliquus*; D-fraction, from *Grifola frondosa*; GLPS polysaccharide fraction from *Ganoderma lucidum*; active hexose correlated compound (AHCC); and many others.[14,46,47]

Other mushroom compounds of therapeutic interest are the secondary metabolites, such as lectins, terpenoids, alkaloids, antibiotics, and metal chelating agents, which are also important for the immune function of the organism. MMs also contain a number of enzymes, such as laccase, superoxide dismutase, glucose oxidase, and peroxidase. It has been shown that enzyme therapy plays an important role in cancer treatment, preventing oxidative stress and inhibiting cell growth.[46,99]

It has been documented that fungi produce a huge number of biologically active compounds that not only stimulate the immune system but also modulate specific cellular responses by interfering in particular transduction pathways. For instance, caffeic acid phenethyl ester (CAPE), which specifically inhibits DNA binding of NF-κB and has shown some promising results in human breast cancer MCF-7 cells, was found to be produced by *Agaricus bisporus*, *Marasmius oreades*, *Lentinus edodes*, and *Phellinus linteus*. Also, a methanol extract of *Fomes fomentarius* was reported to inhibit iNOS and COX expression due to the down-regulation of NF-κB binding activity to DNA. Panepoxydone, a compound isolated from *Panus* spp. but also found in *Lentinus crinitus*, interferes with the NF-κB mediated signal by inhibiting the phosphorylation of IκBα. These reports demonstrate the fact that such substances can be used as molecular targets in malignant cells in order to combat cancer. Because low molecular sizes help them to penetrate the cell membrane, these substances have also been classified as low-molecular-weight compounds; among them are lectins, lactones, terpenoids, alkaloids, antibiotics, and metal-chelating agents. Many fungal species have already been reported to produce various metabolites capable of modulating different intracellular pathways, thus playing an essential role in cancer treatment.[99–102]

The development of real immunomodulating and anticancer drugs from MM polysaccharides (e.g., Lentinan, Schizophyllan, and Krestin) was hampered by the fact that high-molecular-weight compounds are used. All MM drugs were developed from high-molecular-weight polysaccharides from 100,000 to 0.5 million Da. These compounds cannot be synthesized; therefore, their production is restricted to extraction from fruit bodies, cultured mycelium, or cultured broth. Such an approach imposes high market prices. Today, science should concentrate on the beneficial medicinal effects of low-weight-molecular compounds produced by MMs, i.e., low-molecular-weight secondary metabolites targeting processes such as apoptosis, angiogenesis, metastasis, cell cycle regulation, and signal transduction cascades.[40] West-

ern pharmaceutical companies are more interested in relatively easily synthesized compounds that can be produced for markets.

Historically, the majority of new drugs have been generated from natural products (secondary metabolites). MMs are an unclaimed source for drug discovery. By 1990, about 80% of drugs were either natural products or analogs inspired by them. "Blockbuster drugs" like antibiotics (penicillin, tetracycline, and erytromycin), antiparasitics (avermectin), anti-malarials (quinine, artemisinin), lipid control agents (lovastatin and analogs), immuno-suppressants for organ transplants (cyclosporin, rapamycins), and anticancer drugs (taxol, doxorubicin) revolutionized medicine.[103] Many of these drugs were developed from components found in fungi.

Hahn and Weinberg[104] designed a subway map (Fig. 4) of cancer pathways involving the genetic changes that liberate neoplastic cells from the homeostatic mechanisms that govern normal cell proliferation. Advances in human genetics and molecular and cellular biology have identified a collection of cell phenotypes that are required for malignant transformation. Specific molecular pathways (subway lines) are responsible for determining these behaviors. Although the connections between cancer-cell wiring and function require further investigation (hence the many lines under construction), a better understanding of the molecular circuitry of the cancer cell can now be sketched. Further advances in describing these pathways and their subsequent connections will elucidate the development of molecularly targeted therapies that will benefit the field of oncology.

Modern pharmaceutical trends in preventing cancer include the development of new drugs with:

(a) Growth-factor inhibitors of cancer cells (drugs such as Herceptin, Erbitux, and Terceva). They block a cancer cell's link to critical proteins that help it divide and grow.

(b) Hormone blockers (drugs such as Tamoxifen) keep cells from dividing by binding to estrogen receptors, which are over-expressed in some tumor cells.

(c) Signal blockers working inside a cell. These drugs interrupt communication among enzymes that regulate growth and development.

(d) Angiogenesis inhibitors, e.g., Avastin, which was the first drug to inhibit the formation of new blood vessels around cancer cells, starving them of nourisment.[105]

About 860 cancer drugs are being tested on humans. This number is more than twice the number of experimental drugs for heart disease and stroke combined, nearly twice as many as for AIDS and all other infectious diseases combined, and nearly twice as many as for Alzheimer's and all other neurological diseases combined.[106] Cancer drugs have been the biggest category of drugs in terms of sales worldwide since 2006 and in the US since 2008, according to market research by IMS Health. Today, drug companies see a future in treating cancer. The world's largest pharmaceutical company (Pfizer), for example, was focused on cardiovascular drugs, the cholesterol-lowering agent Lipitor,[107] and the blood pressure reduction pill Norvasc.[106] Recently, Pfizer hired about 1,000 researchers for an all-out effort to develop drugs for cancer, a disease the company once largely ignored. Pfizer has now scaled back on cardiovascular research and has made cancer drugs one of its six focus areas. About 20% of Pfizer's more than $7 billion budget for research and development spending is on cancer research, and 22 of the roughly 100 drugs being tested are anticancer drugs.[106]

Progress in research of MMs must include genomics, proteomics, metabolomics, and systems pharmacology. Studying molecular mechanisms to determine the medicinal effects of MMs should be the focus of new investigations using modern methods in the above approaches.

Another important source for substances of therapeutic interest can be found in the pool of secondary metabolites produced by MMs. These substances can be classified according to five main metabolic sources:[99] amino acid–derived pathways; the shikimic acid pathway for the biosynthesis of aromatic amino acids; the acetate-malonate pathway from acetyl coenzyme A; the mevalonic acid pathway from acetyl coenzyme A, which functions in primary metabolism for the synthesis of sterols; and polysaccharides and peptidopolysaccharides. The polyketide and the mavalonic acid pathways are most often involved, and they produce a greater variety of compounds than the other pathways.

Every effort should be made to find new sources for anticancer drugs using low-molecular-weight secondary metabolites from MMs that can inhibit or trigger specific responses, i.e., activat-

FIGURE 4. A subway map of cancer pathways (subway map designed by Claudia Bentley. Web design by Nick Allin. Edited by Cath Brooksbank and Sandra Clark).

ing or inhibiting NF-κB, inhibiting protein and especially tyrosine kinases, aromatase and sulfatase, matrix metalloproteinases, cyclo-oxygenases, DNA topoisomerases and DNA polymerase, anti-angiogenic substances, etc.[99–102]

Fungal low-molecular-weight compounds directly influencing NF-κB inhibitory effects are: caffeic acid phenethyl ester (CAPE), cordycepin, panepoxydone, and cycloepoxydon. Low-molecular-weight CAPE produced, for example, by *Phellinus linteus* and *Marasmius oreades*, show specific cytotoxicity against tumor cells, show NF-κB inhibitor activity, and can be a candidate for antitumor drugs, especially against breast cancer.[102]

Pharmaceutical companies involved in drug discovery need new sources of natural products.

MMs are the best unclaimed gifts of nature that in a short amount of time can be used in the production of new pharmaceuticals.

Below we will detail the drug discovery pathway for the development of mushroom pharmaceuticals. The drug discovery pathway for mushroom bioactive metabolites includes the following basic steps:[14]

A. Mushroom Cultivation and Biomass Production

First, selected mushroom strains are grown on solid medium (agar-agar, malt extract agar, susloyeast extract agar medium, etc.) in Petri dishes for the production of pure mycelia cultures. Different mushroom species develop mycelium for different periods of time depending on their specific species characteristics and on cultivation conditions such as medium nutrition contents, pH, and temperature. Therefore, according to the main purpose of the study, the best growth conditions for selected strains should be chosen in order to receive high yields of biomass and bioactive substances. Basically, when grown on solid medium, mushroom strains are maintained in an incubator at 27°C till the Petri dish surface is fully covered with fresh mycelium. When the pure mycelia cultures are completely developed and ready for use, the second stage begins, called submerged mycelia cultivation for biomass production. Fresh mycelia, cut into particles, are inoculated in liquid medium (with the same nutrition contents, only lacking agar) and grown on a rotary shaker till the mycelia particles totally fill the medium. This stage varies in the amount of time, according to the species' characteristics. For biomass production, obtained mycelia inoculums are homogenized and transferred into a larger amount of liquid culture medium and then put on a shaker (or in a fermentor) for a specified period of time.

B. Biomass Extraction

The biomass received from each strain should be dried and ground into fine powder. According to the selected mushroom compounds, specific organic solvents for the extraction should be chosen. For instance, substances that are supposed to penetrate the cell membrane have lipid origin and should be extracted with water-insoluble solvents, such as ethyl acetate. In contrast, compounds that are expected to act on the cell surface could be extracted with water-soluble solvents, such as alcohols. Generally, the most common organic solvents used for biomass extraction are methanol, ethyl alcohol, ethyl acetate, diethyl ether, formalin, etc., or some combinations of these. Moreover, the culture liquid could also be extracted (with water-insoluble solvents, e.g., ethyl acetate), assuming that it is very rich in bioactive mushroom secondary metabolites.

C. Screening of Mushroom Extracts

The primary screening of mushroom extracts obtained in the stage mentioned above could be performed using different methods and assays. Depending on the main target of the study, a specific cancer cell line should be selected. One possibility is to apply the luciferase activity reporter assay. Transfected cells carrying a luciferase reporter gene under the control of a specific promoter should be used. In this case, the extracts' effects are evaluated spectrophotometrically according to the exposed light and are calculated as a percent of the reporter inhibition. Another possibility is for extracts to be tested in their ability to affect cancer cell viability using the cell proliferation assay, in which cells are treated with the extracts for a definite period of time, and afterwards the viable cells are counted using the trypan blue exclusion method or MTT and XTT methods. Again, extract activity is calculated as a percent of cancer cell growth inhibition and IC_{50} (the extracts concentration that inhibits cell growth by 50%) can be determined. The primary screening should be carried out at least in duplicate.

Extracts that show reporter inhibition higher than 50% and cancer cell growth inhibition lower than 50% should be considered as active and should be subjected to a second screening in order to confirm their positive effect. For the re-screening procedure, the same methods can be applied, but more concentrations of the selected extracts should be used for more precise evaluation of their effect.

D. Effect of Selected Extracts on a Target of Interest

The levels of target cell proteins responsible for cancer cell growth, proliferation, metastasis, drug resistance, etc., are checked in response to select-

ed extracts. Cancer cells are seeded, treated with extracts in certain concentrations, collected, and lyzed. The proteins of interest can be detected using methods such as Western blot or ELISA. The expression of specific genes involved in the process of tumorigenesis can also be determined using PCR for detecting the expression of target genes, RT-PCR for establishing the approximate level of gene expression, and real-time PCR for determining the exact level of gene expression.

Establishment of the levels of target proteins and genes in accordance with the effect of mushroom extracts will result in the elucidation of their specific mechanism of action on the intracellular level. Afterwards, the most potent extracts that significantly affect the targets of interest will be selected for further evaluation and subjected to chemical fractionation in order to determine the chemical origin of the active moieties.

E. Chemical Fractionation of Selected Extracts

The most active extracts will be fractioned into their component compounds by passing them through a system that retards each compound to a varying degree; a system capable of accomplishing this process is called a chromatograph. The retarding system can be a surface adsorbant such as silica, alumina, cellulose, or charcoal, capable of reversibly adsorbing the compound. Chemical fractionation might be performed by HPLC, a technique for separating proteins and other molecules, TLC, or ion-exchange chromatography. As a result, fractions with exact chemical contents such as protein, polysaccharide, lipid fractions, etc., will be obtained.

F. Elucidation of Active Fractions' (Compounds') Mechanism of Action and Potency

All fractions obtained in the process of extract chemical fractionation will be evaluated for their effect on the targets of interest in order to determine and select the fraction that contains the most active substances. For this purpose, the same methods, e.g., Western Blot, ELISA, cell proliferation assay, PCR, RT-PCR, real-time PCR, etc., can be applied. Positive mushroom fractions that show valuable effects on the proteins of interest will be subjected to additional evaluation with the aim of determining their specific mechanism of action.

G. Effect on Animal Models

Fractions that showed a desired effect in *in vitro* assays will be tested for their *in vivo* effect in animal models. Rats and nude mice (mice that cannot reject tumors or transplants of cells from humans or other animals) can be used for this purpose. Tumors will be induced subcutaneously into the selected laboratory animals and left to progress. Afterwards, the active mushroom fractions (compounds) will be injected into the tumor and their effect on tumor growth and metastasis will be periodically measured. After the animal has been sacrificed, another method is to take the tumor out and to treat it in culture with the mushroom substances in order to determine their direct effect on the tumor cells.

Fractions that demonstrate a significant anticancer activity in *in vivo* assays will be subjected to additional chemical analyses for determining the exact chemical structure of the bioactive compound of interest. Medicinal chemistry (chemical modification of the molecule) can be performed in order to obtain a better and more potent compound. Such a compound will further represent a potential candidate drug.

H. Preclinical Drug Development

The candidate drug is chemically pre-formulated, formulated, and then tested for metabolism, toxicity, bioavailability, and other pharmacological properties; this was traditionally done in animals but increasingly in *in vitro* model assays. If the desired effect is obtained, the product stability is checked, and then the candidate drug can be given to a pilot manufacturing facility.

I. Clinical Drug Development

The final drug development stage includes several phases. First, the manufactured candidate drug needs an application (permission) for approval for clinical trials. When the product is approved for clinical trials, three phases of clinical trials begin: (1) testing the product's safety in volunteers; (2) finding evidence for efficacy in patients and estimation of the appropriate dose; and (3) accurate assessment of clinical values of the candidate drug. Each of these phases includes systematic and extensive records, which are used in obtaining per-

mission to market the product as a drug.[14]

XI. CONCLUDING REMARKS

Edible and MMs represent promising and relatively untapped sources of materials with potential human food and medicinal applications. Therefore, it is time that humans make every endeavor to maximize the use of their agricultural waste resources. We should thus make a concerted effort in research and development in mushroom cultivation. It is also the time that we should lobby the governments and international organizations, as well as industrial circles, reminding them that research and development in mushroom-based industries is not a luxury, but a national necessity. In spite of the many problems that exist in the cultivation of mushrooms, we foresee a new possibility of using mushrooms in a more important and positive role, especially as a reliable source of protein, to enrich human diets, and as MM DSs, to enhance immune responses. This is especially necessary in less developed countries, where the shortage of protein and lack of healthcare are more marked than in most of the developed countries. We also see no reason why both rural and urban areas cannot benefit from this new technology to maximize mushroom production per unit area with minimum costs, thus generating a new cheap source of food protein from agricultural waste materials, and also to initiate mushroom biotechnology in processing mushroom products, thus providing health-enhancing dietary supplements for the people.

The progress achieved during the past two decades in both research and production has clearly documented evidence of the benefits mushrooms have to offer as human food as well as to human healthcare systems. Modern medicines may not always provide the remedies required and MM DSs are likely to prove extremely useful in both alleviating and preventing human disease conditions. Disease prevention is particularly desirable, not only in having positive financial and social impacts, but also in maintaining/improving the quality of life and human dignity. We hope our humble appeal for the promotion of mushroom-based industry worldwide presents a challenge and receives favorable attention and consideration for active implementation.

MM-derived products are neither nutraceuticals (food/functional food) nor pharmaceuticals (drugs), because the active ingredient of most products is not a single, chemically defined compound as used in conventional drug treatments. Therefore, they may be classified as a type of DS or traditional medicine, which is a category between food and drugs.

There are some mushroom-derived studies on MMs that claim that mushroom products, e.g., *Ganoderma lucidum* or *Cordyceps sinensis*, hold the key to the relief of all human illnesses. We think this is over-promoted. Actually there is no known modality of treatment including both orthodox and traditional medicines that will fix everybody every time.

We should not assume that mushroom-derived products can replace modern medicine. We can, however, recognize that generally mushroom-derived products are a special category that can serve a patient better under certain circumstances (e.g., enhance general well-being) and other situations serve to supplement other treatments (e.g., complement modern medicine).

As mentioned in the beginning, mushrooms are used as a health food, as a tonic, and as a medicine (drugs and botanical drugs). A regular intake of mushrooms themselves and mushroom derivatives makes one healthier, fitter, and happier.

They are rich in medicinal properties—in particular, in selenium (antioxidant)—that make one look younger and live longer. What we have from this review is scientific, economic, healthful, and environmental aspects of mushroom production and uses.

We have discussed the legal and regulatory issues involved in introducing and controlling DSs from medicinal mushrooms in different countries, including the United States, the EU, Canada, Australia, New Zealand, Japan, and P.R. China, and guidelines of the World Health Organization. Much attention has highlighted safety and diversity of DS types from culinary-medicinal mushrooms. Most of the MM DSs currently in the marketplace are highly diverse, and there are only a few standard protocols to ensure product quality. There must be thorough analyses, improved quality, and legal control, which will, in turn, increase and maintain consumer confidence and achieve the current and future standards set by national regulatory authorities. We hope that these and future regulations will continue to provide and protect stable

medicinal mushroom products of reliable quality and that culturing biotechnology will continue to be the best and the most progressive technique for obtaining consistent and safe mushroom products.

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